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Scaled projection neural network for quasi-variational inequalities

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ABSTRACT

We propose a scaled projection neural network for solving *quasi-variational inequalities* in which the constraint set $\mathfrak{S}(x) = m(x) + S$ depends on the state through a contraction mapping m that translates a fixed closed convex set S , while the projection is performed with respect to a fixed symmetric positive definite matrix M . The matrix M acts as a continuous-time preconditioner, reshaping the geometry of the projected flow without altering the solution set. We establish that equilibria of the proposed dynamics coincide exactly with solutions of the quasi-variational inequality, and prove well-posedness under Lipschitz regularity of the operator and the translation mapping. A stability analysis based on a weighted Lyapunov function induced by the inverse of M yields an explicit contraction parameter that captures the interplay between the matrix bounds and the contraction properties of the translation, generalizing classical step-size conditions for both fixed-set variational inequalities and quasi-variational inequalities. Under an explicit dominance condition relating the monotonicity modulus, the Lipschitz constants, and the matrix bounds, we prove global exponential convergence on a computable step-size window, and a forward-Euler discretization preserves the certified rate. Numerical experiments across five application domains illustrate the role of M as a preconditioner and exhibit convergence in regimes where the certificate is silent. The framework connects and extends two lines of research that have not been previously combined: Euclidean projected dynamical systems for quasi-variational inequalities and state-dependent matrix projection neural networks for variational inequalities with fixed constraint sets.

1. Introduction

Variational inequality (VI) problems provide a unifying framework for a wide range of equilibrium models arising in optimization, game theory, network analysis, and constrained dynamical systems. Given a nonempty closed convex set $S \subset \mathbb{R}^n$ and a continuous mapping $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$, the classical *variational inequality problem* $VI(F, S)$ consists of finding a point $\bar{x} \in S$ such that

$$\langle F(\bar{x}), y - \bar{x} \rangle \geq 0, \quad \forall y \in S. \quad (1)$$

When the feasible set itself depends on the current state, the problem becomes a quasi-variational inequality, described as follows: Given a set-valued mapping $\mathfrak{S} : \mathbb{R}^n \rightrightarrows \mathbb{R}^n$ such that $\mathfrak{S}(x)$ is a nonempty closed convex set for each $x \in \mathbb{R}^n$, the *quasi-variational inequality problem* $QVI(F, \mathfrak{S})$ consists of finding a point $\bar{x} \in \mathfrak{S}(\bar{x})$ such that

$$\langle F(\bar{x}), y - \bar{x} \rangle \geq 0, \quad \forall y \in \mathfrak{S}(\bar{x}). \quad (2)$$

It was introduced by Bensoussan et al. (1973) in connection with impulse control problems, and further systematically studied in Baiocchi and Capelo (1984), Bensoussan and Lions (1982). The generalized Nash

equilibrium problem (Facchinei & Kanzow, 2010) and several problems from contact mechanics, traffic network equilibria, and differential games (Pang & Stewart, 2008) can also be written in the form of a quasi-variational inequality problem.

In many applications, the state-dependent constraint set $\mathfrak{S}(x)$ has the translatable form

$$\mathfrak{S}(x) = m(x) + S, \quad (3)$$

where $m : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a point-to-point mapping and S is a fixed closed convex set. This structure arises naturally in generalized Nash games where each player's feasible region is shifted by the strategies of others, and in economic models where capacity constraints depend on aggregate demand.

The projection method is one of the earliest methods to solve variational/quasi-variational inequality problems; see, e.g., Ansari et al. (2013b), Facchinei and Pang (2004), Noor (1988) and the references therein.

A parallel line of research introduces *scaled* or *preconditioned* projections. The *skewed projector* $\Pi_{K,A}$, which minimizes the A -norm distance over a closed convex set K for a fixed symmetric positive definite (SPD)

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matrix A , was studied systematically in [Facchinei and Pang \(2004\)](#); its firm nonexpansiveness in the A -norm and Lipschitz continuity with constant $\text{cond}(A) = \|A\| \|A^{-1}\|$ (the condition number of A) form the analytical foundation for scaled projection methods.

In discrete-time optimization, [Bonettini et al. \(2009\)](#) introduced the scaled gradient projection (SGP) method, which incorporates a positive definite scaling matrix into the projection step for constrained optimization. This approach was further developed by [Bonettini and Prato \(2015\)](#), who demonstrated that the scaling provides a better rate of convergence, particularly for ill-conditioned problems.

On the other hand, projection-based dynamical systems and projection neural networks have been widely used as continuous-time approaches for solving variational inequalities. The classical projected dynamical system (PDS), introduced by [Dupuis and Nagurney \(1993\)](#), takes the form

$$\frac{dx}{dt} = \lambda(P_S(x - \alpha F(x)) - x), \quad (4)$$

where P_S denotes the Euclidean projection onto S , and $\lambda > 0$, $\alpha > 0$ are parameters. Equilibria of (4) coincide with solutions of $\text{VI}(F, S)$, and Lyapunov-based stability analysis guarantees convergence under monotonicity-type assumptions on F . A full treatment is given in [Nagurney and Zhang \(1996\)](#); extensions to Hilbert spaces and evolutionary settings were developed in [Cojocaru et al. \(2005\)](#).

The use of analog circuit dynamics for optimization goes back to [Hopfield and Tank \(1985\)](#), [Tank and Hopfield \(1986\)](#) and to the canonical nonlinear programming circuit of [Kennedy and Chua \(1988\)](#). Projection neural networks (PNNs) realize the same dynamics in a neural or hardware-oriented architecture. [Xia and Wang \(2004\)](#) proposed a general projection neural network (GPNN) with a diagonal scaling matrix Λ multiplying the right-hand side and established global convergence under G -monotonicity. [Hu and Wang \(2006\)](#) extended the stability analysis to pseudomonotone operators, and later [Hu and Wang \(2007\)](#) introduced a matrix preconditioner W for general variational inequalities. [Gao et al. \(2005\)](#) handled nonlinear constraints via KKT-based reformulations. More recently, [Yang et al. \(2025\)](#) achieved fixed-time convergence under strong pseudomonotonicity, and [Allibhoy and Cortés \(2025\)](#) developed safe monotone flows with forward-invariance guarantees. Neurodynamic optimization remains an active area well beyond variational inequalities, with recent continuous-time models for portfolio allocation ([Cao et al., 2025c](#)) and k -winner-take-all selection ([Cao et al., 2025b](#)). A key observation across all these works is that the projection operator $P_S(\cdot)$ is always Euclidean: while some papers scale the right-hand side of the ODE (via Λ or W), the projection itself is computed with respect to the standard inner product. The underlying geometry of the projection step is therefore not adapted to the problem structure.

The extension of projection-based dynamics from VI to quasi-variational inequalities — where the constraint set varies with the state — has received comparatively little attention. The most closely related work is [Noor \(2003\)](#), who proposed implicit projected dynamical systems for QVIs of the form

$$\frac{du}{dt} = \lambda\{P_{K(u)}[u - \rho A(u)] - u\},$$

where $K(u)$ is a state-dependent closed convex set. Under pseudomonotonicity of A and a Lipschitz condition on the projection mapping ($\|P_{K(u)}w - P_{K(v)}w\| \leq \nu\|u - v\|$), he established global asymptotic stability, and exponential stability under Lipschitz continuity. For the dynamical system for extended general nonlinear quasi-variational inequalities, we refer to [Ansari et al. \(2013a\)](#).

Motivated by the above facts, recent work ([Alshahrani, 2026](#)) initiated a systematic study of projection neural networks with a *state-dependent* matrix $\mathbb{M}(x)$, proposing the dynamics

$$\frac{dx}{dt} = \lambda[P_{S, \mathbb{M}^{-1}(x)}(x - \alpha \mathbb{M}(x)F(x)) - x], \quad (5)$$

where $P_{S, \mathbb{M}^{-1}(x)}$ denotes the projection with respect to the matrix $\mathbb{M}^{-1}(x)$. That work introduced the state-dependent-matrix framework

for variational inequalities with a *fixed* feasible set, and studied its well-posedness, equilibrium characterization, and stability under uniform boundedness and Lipschitz regularity of the matrix mapping ([Alshahrani, 2026](#)).

The present paper occupies a position at the intersection of three research streams that have not been previously connected: (i) QVI dynamics ([Noor, 2003](#)) with state-dependent constraint sets but Euclidean projection; (ii) scaled projection methods ([Alshahrani, 2026](#); [Bonettini et al., 2009](#)) with variable metric projection but fixed constraint sets; and (iii) projection neural networks ([Hu & Wang, 2006](#); [Xia & Wang, 2004](#)) with continuous-time dynamics but Euclidean projection and fixed sets. No existing framework incorporates a general scaling matrix into a projection neural network for QVI.

The present work addresses this gap by proposing a *scaled projection neural network for quasi-variational inequalities*. We keep the matrix M fixed (a symmetric positive definite matrix) and allow the constraint set to be state-dependent via the translative form (3). Specifically, we study the dynamical system

$$\frac{dx}{dt} = \lambda[P_{\mathfrak{S}(x), M^{-1}}(x - \alpha M F(x)) - x], \quad (6)$$

where $P_{\mathfrak{S}(x), M^{-1}}$ denotes the metric projection onto $\mathfrak{S}(x) = m(x) + S$ with respect to the inner product induced by M^{-1} . When $M = I$ and $m(x) \equiv 0$, the system reduces to the classical projected dynamical system (4).

The analytical challenge in (6) is that the feasible set moves with the state through $m(x)$, so the projection operator depends on x through both its argument $x - \alpha M F(x)$ and the set $\mathfrak{S}(x)$ onto which it projects. We now make the relation to [Alshahrani \(2026\)](#) precise. The projection architecture — a fixed-point map built from a metric projection of a forward step — and the variational-characterization argument are inherited from that analysis, which treats a fixed set with a state-dependent matrix. What is new is the analysis of the moving set. The translation identity below gives an exact decomposition of the moving-set projection. The set motion contributes the outer term $m(x) - m(y)$ to the Lipschitz and contraction estimates, and the mixed product $\langle m(x) - m(y), F(x) - F(y) \rangle$ to the key contraction estimate; the product enters that estimate through the effective modulus μ_{eff} and is handled naturally by the M^{-1} -weighted analysis of [Section 4](#). The resulting constants and guarantees are globally explicit in the problem data: the Lipschitz constant of the fixed-point map, global well-posedness, the closed-form contraction factor ρ_M with its admissible step-size window, and a discrete-time convergence guarantee. Conversely, the state-dependent matrix $\mathbb{M}(x)$ of [Alshahrani \(2026\)](#) is held fixed in this paper. The fully coupled case remains open ([Section 6](#)).

A key structural property exploited throughout our analysis is the *translation identity*:

$$P_{m(x)+S, M^{-1}}(z) = m(x) + P_{S, M^{-1}}(z - m(x)), \quad (7)$$

which decomposes the projection onto a moving set into a translation plus a projection onto the fixed set S . This identity allows us to isolate the contribution of the set variation $m(x)$ from the projection mechanics, leading to clean Lipschitz estimates.

Contributions. The main contributions of this paper are as follows:

- We introduce a scaled projection neural network (6) for quasi-variational inequality problems with a fixed positive definite matrix M and a state-dependent constraint set $\mathfrak{S}(x) = m(x) + S$.
- We establish an equilibrium characterization showing that the stationary points of (6) coincide exactly with the solutions of $\text{QVI}(F, \mathfrak{S})$.
- We prove well-posedness (existence and uniqueness of trajectories) under Lipschitz regularity of both F and the translation mapping m .
- We develop a Lyapunov-based stability analysis using an M^{-1} -weighted Lyapunov function, establishing Lyapunov stability and global exponential convergence on an explicit step-size window under the quantitative conditions of (24). We also give a forward-Euler discretization that preserves the exponential rate. The analysis explicitly accounts for the interaction between the fixed matrix

M , the moving constraint set, and the contraction properties of m , yielding an explicit step-size condition that generalizes classical results for both fixed-set variational inequalities and Euclidean quasi-variational inequality problems.

- We present nine numerical experiments across five application domains — including a nonlinear benchmark inspired by the QVI dynamical systems literature, an implicit obstacle problem from contact mechanics, and a generalized Nash equilibrium from industrial economics — that illustrate the certified convergence behavior and probe its conservatism. The dynamics converge for step sizes exceeding the reference scale of Section 5.2 by several orders of magnitude. Larger set motion K_m degrades convergence only in the weakly monotone and non-monotone cases tested.

The remainder of the paper is organized as follows. Section 2 introduces notation, the metric projection, and the standing assumptions that underpin the analysis. Section 3 establishes well-posedness and the equilibrium characterization. Section 4 develops the stability analysis: Lyapunov stability via a weighted-norm approach, global exponential convergence on an explicit step-size window, and a rate-preserving forward-Euler discretization. Section 5 presents numerical experiments, and Section 6 offers concluding remarks.

2. Formulation and notation

We begin by introducing the notation, the metric projection, and the standing assumptions that underpin the analysis.

Let $S \subset \mathbb{R}^n$ be a nonempty closed convex set, let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a continuous mapping, and let $m : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a continuous mapping defining the state-dependent constraint set $\mathfrak{S}(x) = m(x) + S$. Throughout the paper, $\langle \cdot, \cdot \rangle$ denotes the standard Euclidean inner product on $\mathbb{R}^n$ with associated norm $\|x\| = \sqrt{\langle x, x \rangle}$. We fix a symmetric positive definite (SPD) matrix $M \in \mathbb{R}^{n \times n}$, which induces the inner product and norm $\langle x, y \rangle_M := x^T M y$, $\|x\|_M := \sqrt{\langle x, x \rangle_M}$.

The central object in our construction is the *metric projection* (or *skewed projector* Facchinei & Pang, 2004) onto S with respect to the matrix M^{-1} :

$$P_{S, M^{-1}}(y) := \operatorname{argmin}_{z \in S} \frac{1}{2} \|z - y\|_{M^{-1}}^2 = \operatorname{argmin}_{z \in S} \frac{1}{2} (z - y)^T M^{-1} (z - y). \quad (8)$$

Since M^{-1} is SPD and S is closed and convex, the minimizer exists and is unique. It admits the following variational characterization.

Lemma 1 (Variational characterization Facchinei & Pang, 2004, Eq. (1.5.10)). *For given $u \in \mathbb{R}^n$, a point $z \in S$ satisfies $z = P_{S, M^{-1}}(u)$ if and only if*

$$(z - u)^T M^{-1} (y - z) \geq 0, \quad \forall y \in S. \quad (9)$$

When the feasible set is translated by a vector $a \in \mathbb{R}^n$, the metric projection decomposes cleanly.

Lemma 2 (Translation identity). *For any $a \in \mathbb{R}^n$ and $z \in \mathbb{R}^n$,*

$$P_{a+S, M^{-1}}(z) = a + P_{S, M^{-1}}(z - a).$$

Proof. Substituting $w = v - a$ with $v \in a + S$ (so $w \in S$) into the definition (8) gives

$$\begin{aligned} & \operatorname{argmin}_{v \in a+S} \frac{1}{2} (v - z)^T M^{-1} (v - z) \\ &= a + \operatorname{argmin}_{w \in S} \frac{1}{2} (w - (z - a))^T M^{-1} (w - (z - a)). \end{aligned}$$

□

Remark 1 (Computing the metric projection). The translation identity has a computational reading: evaluating the projection onto the moving set costs one evaluation of $m(x)$ plus a single projection onto the fixed set S , so the state dependence of the constraint adds no projection overhead along the dynamics. For $P_{S, M^{-1}}$, consider box, orthant, or

one-sided bound constraints with *diagonal* M . These are the sets in all experiments of Section 5 except the Nash game of Experiment 6, whose set is a box intersected with one shared linear constraint. In this setting, the minimization (8) is separable and reduces to the componentwise clamp, at cost $O(n)$. This is also the configuration the experiments recommend (Experiment 8). For a general dense SPD M on a box, the projection is a bound-constrained quadratic program. Since M is fixed, the Hessian and constraints of this program are fixed along the trajectory; only the center z changes. The subproblem, small at the dimensions of our experiments, is therefore warm-startable. We solve it by projected coordinate descent (Section 5). A change of variables does not reduce it to a clamp, since the box becomes oblique. See Bauschke and Combettes (2017), Beck (2017) for standard projection computations.

The metric projection also enjoys strong regularity properties inherited from the SPD structure of M^{-1} .

Lemma 3 (Firm nonexpansiveness Facchinei & Pang, 2004, Theorem 1.5.5). *Let $p_u := P_{S, M^{-1}}(u)$ and $p_v := P_{S, M^{-1}}(v)$. The projection $P_{S, M^{-1}}$ is firmly nonexpansive with respect to the M^{-1} -inner product: that is, for all $u, v \in \mathbb{R}^n$,*

$$(p_u - p_v)^T M^{-1} (u - v) \geq \|p_u - p_v\|_{M^{-1}}^2. \quad (10)$$

In particular, $P_{S, M^{-1}}$ is nonexpansive in the M^{-1} -norm:

$$\|p_u - p_v\|_{M^{-1}} \leq \|u - v\|_{M^{-1}}, \quad \forall u, v \in \mathbb{R}^n,$$

and Lipschitz continuous in the Euclidean norm with constant $\kappa := \operatorname{cond}(M)$:

$$\|p_u - p_v\| \leq \kappa \|u - v\|, \quad \forall u, v \in \mathbb{R}^n.$$

We now state the standing assumptions that will be maintained throughout the analysis. The first concerns the matrix M itself.

Assumption 1 (Matrix boundedness). The fixed SPD matrix M satisfies $\|M\| \leq L$ and $\|M^{-1}\| \leq L$ for some constant $L \geq 1$, where $\|\cdot\|$ denotes the spectral norm.

Assumption 1 implies that the eigenvalues of M lie in $[L^{-1}, L]$, so $\kappa = \operatorname{cond}(M) \leq L^2$, and all induced norms are uniformly equivalent to the Euclidean norm:

$$\begin{aligned} \frac{1}{\sqrt{L}} \|z\| &\leq \|z\|_M \leq \sqrt{L} \|z\|, \\ \frac{1}{\sqrt{L}} \|z\| &\leq \|z\|_{M^{-1}} \leq \sqrt{L} \|z\|, \quad \forall z \in \mathbb{R}^n. \end{aligned}$$

The next two assumptions govern the data of the quasi-variational inequality. They are stated on all of $\mathbb{R}^n$: the trajectories of the dynamics introduced below, and the iterates of the discrete scheme of Section 4.3, need not remain in S .

Assumption 2 (Translation regularity). The mapping $m : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a contraction: that is, there exists $K_m \in [0, 1)$ such that

$$\|m(x) - m(y)\| \leq K_m \|x - y\|, \quad \forall x, y \in \mathbb{R}^n.$$

The contraction condition $K_m < 1$ is standard in the theory of quasi-variational inequality problems (Chan & Pang, 1982; Noor, 1988); it ensures that the set-valued mapping $x \mapsto \mathfrak{S}(x)$ does not shift too rapidly, and it underlies the existence and stability theory developed below. A natural special case is the affine translation $m(x) = Ax + b$ with $\|A\| < 1$, which arises frequently in generalized Nash equilibrium problems (Facchinei & Kanzow, 2010).

Assumption 3 (Lipschitz continuity). The mapping $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is Lipschitz continuous with constant $K > 0$: that is,

$$\|F(x) - F(y)\| \leq K \|x - y\|, \quad \forall x, y \in \mathbb{R}^n.$$

The stability analysis will require monotonicity conditions on the operator F , which we state here for completeness.

Assumption 4 (Monotonicity of F). The mapping $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is monotone: that is,

$$\langle F(x) - F(y), x - y \rangle \geq 0, \quad \forall x, y \in \mathbb{R}^n.$$

Assumption 5 (Strong monotonicity of F). The mapping $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is strongly monotone with modulus $\mu > 0$: that is,

$$\langle F(x) - F(y), x - y \rangle \geq \mu \|x - y\|^2, \quad \forall x, y \in \mathbb{R}^n.$$

Assumption 5 implies Assumption 4. Under Assumptions 1–3, if S is additionally compact, the quasi-variational inequality $\text{QVI}(F, \mathfrak{S})$ admits at least one solution. Indeed, the map T of (13) below satisfies $T(x) \in m(x) + S$, so $\|T(x)\| \leq K_m \|x\| + \|m(0)\| + \max_{s \in S} \|s\|$. Since $K_m < 1$, T maps a sufficiently large closed ball into itself. As T is also continuous (Lemma 6 below), Brouwer's fixed-point theorem yields a fixed point of T , which solves the QVI by Proposition 1 below. For general existence theory see Chan and Pang (1982), Facchinei and Pang (2004). In contrast with fixed-set variational inequalities, strong monotonicity alone does not rule out multiple solutions when the constraint set moves; uniqueness is restored once the monotonicity modulus dominates the set motion.

Lemma 4 (Uniqueness). Suppose Assumptions 2, 3, and 5 hold. If $\mu > K_m K$, then $\text{QVI}(F, \mathfrak{S})$ has at most one solution.

Proof. Let $\bar{x}$ and $\bar{y}$ solve $\text{QVI}(F, \mathfrak{S})$. Since $\bar{y} - m(\bar{y}) \in S$, the point $\bar{y} - m(\bar{y}) + m(\bar{x})$ belongs to $\mathfrak{S}(\bar{x})$, so testing (2) at $\bar{x}$ with this point gives

$$\langle F(\bar{x}), \bar{y} - \bar{x} + m(\bar{x}) - m(\bar{y}) \rangle \geq 0.$$

Symmetrically, $\langle F(\bar{y}), \bar{x} - \bar{y} + m(\bar{y}) - m(\bar{x}) \rangle \geq 0$. Adding the two inequalities and rearranging,

$$\langle F(\bar{x}) - F(\bar{y}), \bar{x} - \bar{y} \rangle \leq \langle F(\bar{x}) - F(\bar{y}), m(\bar{x}) - m(\bar{y}) \rangle.$$

By Assumption 5 the left-hand side is at least $\mu \|\bar{x} - \bar{y}\|^2$, while by the Cauchy-Schwarz inequality and Assumptions 3 and 2 the right-hand side is at most $K_m K \|\bar{x} - \bar{y}\|^2$. Since $\mu > K_m K$, this forces $\bar{x} = \bar{y}$. $\square$

With these ingredients in place, motivated by the projection neural network framework and the scaled projection approach of Alshahrani (2026), we introduce the following dynamical system:

$$\frac{dx}{dt} = \lambda [P_{\mathfrak{S}(x), M^{-1}}(x - \alpha M F(x)) - x], \quad (11)$$

where $\lambda > 0$ and $\alpha > 0$ are given parameters. Using the translation identity (Lemma 2), we write (11) equivalently as

$$\frac{dx}{dt} = \lambda [m(x) + P_{S, M^{-1}}(x - m(x) - \alpha M F(x)) - x]. \quad (12)$$

Define the state-dependent fixed-point map

$$\begin{aligned} T(x) &:= P_{\mathfrak{S}(x), M^{-1}}(x - \alpha M F(x)) \\ &= m(x) + P_{S, M^{-1}}(x - m(x) - \alpha M F(x)), \end{aligned} \quad (13)$$

so that the dynamics become

$$\frac{dx}{dt} = \lambda (T(x) - x). \quad (14)$$

A point $\bar{x} \in \mathbb{R}^n$ is an equilibrium of (14) if and only if $T(\bar{x}) = \bar{x}$, i.e.,

$$\bar{x} = P_{\mathfrak{S}(\bar{x}), M^{-1}}(\bar{x} - \alpha M F(\bar{x})). \quad (15)$$

Remark 2 (Network interpretation). The dynamics (14) form a recurrent feedback interconnection of three blocks — the operator block F , the translation block m , and a projection block $P_{S, M^{-1}}$ — closed through a bank of n integrators: the architecture of projection neural networks (Xia et al., 2002; Xia & Wang, 2004). For box constraints and diagonal M (Remark 1), the projection block acts componentwise as a saturation nonlinearity, the classical piecewise-linear activation of analog circuit implementations (Liang & Si, 2001). When F and m are additionally affine, this recommended configuration is a standard single-layer recurrent analog architecture in the lineage of Hopfield and Tank (1985), Kennedy and Chua (1988). For nonlinear F or m , the operator blocks are general nonlinear units. For nondiagonal M the projection block is a coupled subproblem, realizable as an inner recurrent network solving the associated quadratic program (Xia & Wang, 2000), at the

cost of additional internal states whose equilibration must be fast relative to the outer dynamics. The forward-Euler scheme of Section 4.3 is the corresponding discrete-time network. Continuous-time neural dynamics for constrained optimization is an active research direction; see, e.g., Cao et al. (2025a) for a recent model for constrained bi-objective quadratic programming.

3. Well-posedness and equilibria

We now establish basic properties of the dynamical system (11): the vector field is Lipschitz continuous, trajectories exist and are unique, and equilibria correspond exactly to the solutions of $\text{QVI}(F, \mathfrak{S})$. Define the vector field $\Phi(x) := T(x) - x$ for $x \in \mathbb{R}^n$, so that the dynamics read $\frac{dx}{dt} = \lambda \Phi(x)$.

Lemma 5 (Lipschitz continuity of the forward step). Define $G(x) := x - m(x) - \alpha M F(x)$. Under Assumptions 1–3,

$$\|G(x) - G(y)\| \leq (1 + K_m + \alpha LK) \|x - y\|, \quad \forall x, y \in \mathbb{R}^n.$$

Proof. For $x, y \in \mathbb{R}^n$,

$$G(x) - G(y) = (x - y) - (m(x) - m(y)) - \alpha M(F(x) - F(y)).$$

By the triangle inequality and Assumptions 1–3, we have

$$\begin{aligned} \|G(x) - G(y)\| &\leq \|x - y\| + \|m(x) - m(y)\| \\ &\quad + \alpha \|M\| \|F(x) - F(y)\| \\ &\leq \|x - y\| + K_m \|x - y\| + \alpha LK \|x - y\| \\ &= (1 + K_m + \alpha LK) \|x - y\|. \end{aligned}$$

$\square$

Lemma 6 (Lipschitz continuity of T). Under Assumptions 1–3, the map T defined by (13) is Lipschitz continuous on $\mathbb{R}^n$:

$$\|T(x) - T(y)\| \leq (K_m + \kappa(1 + K_m + \alpha LK)) \|x - y\|$$

for all $x, y \in \mathbb{R}^n$.

Proof. Write

$$\begin{aligned} T(x) - T(y) &= (m(x) - m(y)) \\ &\quad + (P_{S, M^{-1}}(G(x)) - P_{S, M^{-1}}(G(y))). \end{aligned}$$

By Lemma 3, the projection $P_{S, M^{-1}}$ is Lipschitz continuous in the Euclidean norm with constant $\kappa = \text{cond}(M)$:

$$\|P_{S, M^{-1}}(u) - P_{S, M^{-1}}(v)\| \leq \kappa \|u - v\|, \quad \forall u, v \in \mathbb{R}^n.$$

Therefore, using Assumption 2 and Lemma 5, we have

$$\begin{aligned} \|T(x) - T(y)\| &\leq K_m \|x - y\| + \kappa \|G(x) - G(y)\| \\ &\leq K_m \|x - y\| + \kappa(1 + K_m + \alpha LK) \|x - y\| \\ &= (K_m + \kappa(1 + K_m + \alpha LK)) \|x - y\|. \end{aligned}$$

$\square$

Lemma 7 (Existence and uniqueness of solutions). Under Assumptions 1–3, for every initial condition $x(0) \in \mathbb{R}^n$, there exists a unique solution $x(\cdot)$ of (11) on $[0, \infty)$.

Proof. By Lemma 6, the map T defined by (13) is Lipschitz continuous on $\mathbb{R}^n$. Hence $\Phi := T - \text{id}$, where id denotes the identity mapping, is globally Lipschitz on $\mathbb{R}^n$. By the Picard-Lindelöf theorem, there exists a unique solution on a maximal interval $[0, T_{\max})$. Since Φ is Lipschitz, $\|\Phi(x)\| \leq \|\Phi(0)\| + L_\Phi \|x\|$ for all $x \in \mathbb{R}^n$, so the solution cannot blow up in finite time and $T_{\max} = \infty$. $\square$

Having established that trajectories are well-defined, we now show that the equilibria of (11) are precisely the solutions of the quasi-variational inequality problem.

Proposition 1 (Equilibrium–solution equivalence). *A point $\bar{x} \in \mathbb{R}^n$ is an equilibrium of (11) if and only if $\bar{x}$ solves QVI($F, \mathfrak{S}$). In either case, $\bar{x} \in \mathfrak{S}(\bar{x})$, so feasibility is automatic.*

Proof. By definition, $\bar{x}$ is an equilibrium of (11) if and only if

$$\bar{x} = P_{\mathfrak{S}(\bar{x}), M^{-1}}(\bar{x} - \alpha M F(\bar{x})).$$

In particular, $\bar{x} \in \mathfrak{S}(\bar{x})$. Using the variational characterization (9) with $u = \bar{x} - \alpha M F(\bar{x})$ and $z = \bar{x}$, we have

$$(\bar{x} - (\bar{x} - \alpha M F(\bar{x})))^\top M^{-1}(y - \bar{x}) \geq 0, \quad \forall y \in \mathfrak{S}(\bar{x}).$$

Since M is symmetric, $M^\top = M$, and thus

$$\alpha(M F(\bar{x}))^\top M^{-1}(y - \bar{x}) = \alpha F(\bar{x})^\top (y - \bar{x}) \geq 0, \quad \forall y \in \mathfrak{S}(\bar{x}).$$

Dividing by $\alpha > 0$ yields $\langle F(\bar{x}), y - \bar{x} \rangle \geq 0$ for all $y \in \mathfrak{S}(\bar{x})$, that is, $\bar{x}$ is a solution of the QVI (2). The converse follows by reversing the argument. $\square$

For later use in the stability analysis, we define the projection residual as follows:

$$r(x) := \frac{1}{\alpha}(x - P_{\mathfrak{S}(x), M^{-1}}(x - \alpha M F(x))) = \frac{1}{\alpha}(x - T(x)). \quad (16)$$

By Proposition 1, $r(x) = 0$ if and only if x solves QVI($F, \mathfrak{S}$). The norm of this residual vector is a classical merit function for quasi-variational inequalities: under strong monotonicity and a Lipschitz condition on the projection mapping, it provides two-sided error bounds on the distance to the solution (Noor, 2007).

4. Stability analysis

We develop the stability theory for the dynamical system (11) in four stages: Lyapunov stability via a weighted-norm analysis, global exponential convergence on an explicit step-size window when strong monotonicity dominates the set motion in the sense of (24), a forward-Euler discretization that preserves the exponential rate, and partial results delimiting what holds beyond the certified window. Throughout this section, $\bar{x}$ denotes a solution of QVI($F, \mathfrak{S}$), equivalently a fixed point of T .

4.1. Lyapunov stability

A natural first approach to Lyapunov stability uses the Euclidean function $V(x) = \frac{1}{2}\|x - \bar{x}\|^2$. Its derivative along trajectories of (14) satisfies $\frac{dV}{dt} = \lambda \langle x - \bar{x}, T(x) - x \rangle \leq 0$ whenever T is nonexpansive in the Euclidean norm. However, the Lipschitz bound from Lemma 6 gives $\|T(x) - T(y)\| \leq (K_m + \kappa(1 + K_m + \alpha LK))\|x - y\|$, and even as $\alpha \rightarrow 0$ the constant $K_m + \kappa(1 + K_m)$ satisfies $K_m + \kappa(1 + K_m) \geq 1 + 2K_m > 1$ for every quasi-variational inequality ($K_m > 0$), since $\kappa = \text{cond}(M) \geq 1$. Thus, whenever $K_m > 0$, the Euclidean-norm bound cannot establish that T is nonexpansive. This motivates working in the M^{-1} -norm, where the projection is firmly nonexpansive without incurring the condition number factor κ .

Consider the M^{-1} -weighted Lyapunov function

$$V_M(x) := \frac{1}{2}\|x - \bar{x}\|_{M^{-1}}^2. \quad (17)$$

Its derivative along trajectories of (14) is

$$\frac{dV_M}{dt} = \lambda[\langle x - \bar{x}, T(x) - \bar{x} \rangle_{M^{-1}} - \|x - \bar{x}\|_{M^{-1}}^2]. \quad (18)$$

We first establish a Lipschitz estimate for T in the M^{-1} -norm with an explicit constant ρ_M , by decomposing differences of T and exploiting the nonexpansiveness of $P_{S, M^{-1}}$.

Lemma 8 (Lipschitz estimate for T in the M^{-1} -norm). *Suppose Assumptions 1–3 hold and F is monotone (Assumption 4). For $x, y \in \mathbb{R}^n$, write $p_x = P_{S, M^{-1}}(G(x))$ and $p_y = P_{S, M^{-1}}(G(y))$, with $G(x) = x - m(x) - \alpha M F(x)$. Then*

$$\|p_x - p_y\|_{M^{-1}}^2$$

$$\leq [(1 + LK_m)^2 - 2\alpha\mu_{\text{eff}} + \alpha^2 L^2 K^2] \|x - y\|_{M^{-1}}^2, \quad (19)$$

where $\mu_{\text{eff}} := L^{-1}\mu - K_m K L$ when F is strongly monotone with modulus μ (Assumption 5), and $\mu_{\text{eff}} := -K_m K L$ under mere monotonicity. Consequently, the map T of (13) satisfies

$$\|T(x) - T(y)\|_{M^{-1}} \leq \rho_M \|x - y\|_{M^{-1}}, \quad \forall x, y \in \mathbb{R}^n, \quad (20)$$

where

$$\rho_M := LK_m + \sqrt{(1 + LK_m)^2 - 2\alpha\mu_{\text{eff}} + \alpha^2 L^2 K^2}. \quad (21)$$

Proof. By the nonexpansiveness of $P_{S, M^{-1}}$ in the M^{-1} -norm, we have

$$\|p_x - p_y\|_{M^{-1}}^2 \leq \|G(x) - G(y)\|_{M^{-1}}^2.$$

Write $h = x - y$, $\delta m = m(x) - m(y)$, $\delta F = F(x) - F(y)$. Then

$$G(x) - G(y) = (h - \delta m) - \alpha M \delta F.$$

Expanding the M^{-1} -norm and using $\langle a, Mb \rangle_{M^{-1}} = a^\top b$, we obtain

$$\begin{aligned} \|G(x) - G(y)\|_{M^{-1}}^2 &= \underbrace{\|h - \delta m\|_{M^{-1}}^2}_{(I)} - 2\alpha \underbrace{(h - \delta m)^\top \delta F}_{(II)} \\ &\quad + \alpha^2 \underbrace{\delta F^\top M \delta F}_{(III)}. \end{aligned} \quad (22)$$

For term (I), by the triangle inequality in the M^{-1} -norm and $\|\delta m\|_{M^{-1}} \leq \sqrt{L} \|\delta m\| \leq \sqrt{L} K_m \|h\| \leq LK_m \|h\|_{M^{-1}}$, we have

$$(I) \leq (1 + LK_m)^2 \|h\|_{M^{-1}}^2.$$

For term (II), under monotonicity of F , we have $h^\top \delta F \geq 0$; while under strong monotonicity, $h^\top \delta F \geq \mu \|h\|^2$, hence $h^\top \delta F \geq \mu L^{-1} \|h\|_{M^{-1}}^2$. Also, $|\delta m^\top \delta F| \leq K_m K \|h\|^2$, which gives $|\delta m^\top \delta F| \leq K_m K L \|h\|_{M^{-1}}^2$. Therefore

$$(II) \geq (\mu L^{-1} - K_m K L) \|h\|_{M^{-1}}^2 =: \mu_{\text{eff}} \|h\|_{M^{-1}}^2.$$

(Set $\mu = 0$ for the merely monotone case.)

For term (III), (III) $\leq \|M\| \|\delta F\|^2 \leq LK^2 \|h\|^2 \leq L^2 K^2 \|h\|_{M^{-1}}^2$.

Substituting the bounds on (I)–(III) into (22) yields

$$\begin{aligned} \|G(x) - G(y)\|_{M^{-1}}^2 &\leq [(1 + LK_m)^2 - 2\alpha\mu_{\text{eff}} + \alpha^2 L^2 K^2] \|h\|_{M^{-1}}^2. \end{aligned} \quad (23)$$

Combining the first display with (23) proves (19). Finally, by (13), $T(x) - T(y) = \delta m + (p_x - p_y)$. The triangle inequality in the M^{-1} -norm, the bound $\|\delta m\|_{M^{-1}} \leq LK_m \|h\|_{M^{-1}}$ (as in term (I)), and (19) give (20). $\square$

Theorem 1 (Lyapunov stability via weighted norm). *Under the hypotheses of Lemma 8, let $\bar{x}$ be a solution of QVI($F, \mathfrak{S}$) and define $V_M(x) = \frac{1}{2}\|x - \bar{x}\|_{M^{-1}}^2$. Then along trajectories of (14):*

$$\frac{dV_M}{dt} \leq \lambda(\rho_M - 1)\|x - \bar{x}\|_{M^{-1}}^2 = -2\lambda(1 - \rho_M)V_M(x),$$

with ρ_M given by (21). In particular, $\bar{x}$ is Lyapunov stable with respect to V_M whenever $\rho_M \leq 1$. When $\rho_M < 1$, the decay $\frac{dV_M}{dt} \leq -2\lambda(1 - \rho_M)V_M(x)$ yields exponential convergence $\|x(t) - \bar{x}\|_{M^{-1}} \leq \|x(0) - \bar{x}\|_{M^{-1}} e^{-\lambda(1 - \rho_M)t}$ via Gronwall's inequality, from every initial state $x(0) \in \mathbb{R}^n$.

Proof. Since $\bar{x}$ is a fixed point of T (Proposition 1), Lemma 8 with $y = \bar{x}$ gives $\|T(x) - \bar{x}\|_{M^{-1}} \leq \rho_M \|x - \bar{x}\|_{M^{-1}}$. By the Cauchy–Schwarz inequality in the M^{-1} -inner product,

$$\langle x - \bar{x}, T(x) - \bar{x} \rangle_{M^{-1}} \leq \rho_M \|x - \bar{x}\|_{M^{-1}}^2.$$

Substituting into (18) yields $\frac{dV_M}{dt} \leq \lambda(\rho_M - 1)\|x - \bar{x}\|_{M^{-1}}^2$. $\square$

Remark 3. The condition $\rho_M < 1$ provides an explicit step-size requirement. When F is strongly monotone ($\mu > 0$) and $\mu_{\text{eff}} > 0$, choosing $\alpha > 0$ to minimize the expression under the square root in (21) yields $\alpha^* = \mu_{\text{eff}} / (L^2 K^2)$, giving

$$\rho_M = LK_m + \sqrt{(1 + LK_m)^2 - \mu_{\text{eff}}^2 / (L^2 K^2)}.$$

This is strictly less than 1 precisely when $LK_m < 1$ and $\mu_{\text{eff}} > 2LK\sqrt{LK_m}$; [Corollary 1](#) below makes the resulting admissible step-size window explicit. When $m \equiv 0$ and $M = I$, we recover $\rho_M = \sqrt{1 - 2\alpha\mu + \alpha^2 K^2}$, the classical contraction constant for projected dynamical systems.

Remark 4 (Mere monotonicity). Under mere monotonicity ($\mu = 0$), the effective constant $\mu_{\text{eff}} = -K_m K L < 0$ for every quasi-variational inequality ($K_m > 0$). Consequently, $\rho_M > 1$ for all $\alpha > 0$, and [Theorem 1](#) does not guarantee Lyapunov stability. The limitation is genuine rather than an artifact of the proof, and it is present already in the fixed-set case $K_m = 0$. Take $F(x) = Jx$ with $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ skew-symmetric (monotone, $\mu = 0$), $m \equiv 0$, and $S = \mathbb{R}^2$. Then the projection is inactive, the dynamics reduce to $\dot{x} = -\lambda\alpha M J x$, and

$$\frac{d}{dt} \frac{1}{2} \|x\|_{M^{-1}}^2 = -\lambda\alpha x^\top J x = 0.$$

Every nonzero trajectory therefore orbits at constant M^{-1} -distance from the unique solution $\bar{x} = 0$, for every α, λ and every SPD M (the same conservation holds for any skew-symmetric $J \neq 0$ in any dimension). No convergence guarantee under mere monotonicity is therefore possible for the dynamics (11) in general; the $\varepsilon = 0$ row of [Table 3](#) is the empirical footprint of exactly this obstruction. By contrast, the merely monotone operator $F(x) = (x_1, 0)^\top$ of [Section 5.3](#) converges: it is the gradient of a convex function, and such co-coercive flows do not exhibit the rotation that defeats the Lyapunov decay. A convergence theory under co-coercivity is a natural intermediate target. Tikhonov regularization $F_\varepsilon = F + \varepsilon I$ enters the certified window (24) only for ε above an explicit positive threshold. With $A_L := L^2(K_m + 2\sqrt{LK_m})$ and K_0 the Lipschitz constant of F , the conditions $A_L < 1$ and $\varepsilon > A_L K_0 / (1 - A_L)$ suffice; no ε works when $A_L \geq 1$. The vanishing-regularization limit therefore remains open. For related directions we refer to the pseudomonotone PNN analysis of [Hu and Wang \(2006\)](#) (fixed sets), the Euclidean QVI dynamics of [Noor \(2003\)](#), algorithms for strongly monotone QVIs ([Nesterov & Scrimali, 2011](#)), and the QVI-to-VI reduction of [Aussel and Sagratella \(2017\)](#).

4.2. Exponential convergence

We now show that when strong monotonicity dominates the set motion and the matrix bounds in the quantitative sense of (24) below, there is an explicit, nonempty step-size window on which $\rho_M < 1$. On this window, T is a global contraction in the M^{-1} -norm and the flow converges exponentially from every initial state. By [Lemma 4](#), strong monotonicity with $\mu > K_m K$ already guarantees at most one solution. The next result additionally provides existence, identifies the basin of attraction — all of $\mathbb{R}^n$ — and gives a convergence rate in the Euclidean norm.

Corollary 1 (Global exponential stability on an explicit step-size window). *Suppose [Assumptions 1–3](#) and [5](#) hold, and*

$$LK_m < 1, \quad \mu_{\text{eff}} > 2LK\sqrt{LK_m}, \quad (24)$$

the latter being equivalent to $\mu > K_m K L^2 + 2L^2 K \sqrt{LK_m}$. Define

$$\alpha_\pm := \frac{\mu_{\text{eff}} \pm \sqrt{\mu_{\text{eff}}^2 - 4L^3 K^2 K_m}}{L^2 K^2}. \quad (25)$$

Then for every $\alpha \in (\alpha_-, \alpha_+)$:

- (i) $\rho_M < 1$, so T is a global ρ_M -contraction on $(\mathbb{R}^n, \|\cdot\|_{M^{-1}})$, and QVI($F, \mathcal{S}$) has a unique solution $\bar{x}$ — no compactness of S is required;
- (ii) every trajectory $x(\cdot)$ of (11), from every $x(0) \in \mathbb{R}^n$, satisfies

$$\|x(t) - \bar{x}\| \leq \sqrt{\kappa} \|x(0) - \bar{x}\| e^{-\lambda(1-\rho_M)t}, \quad \forall t \geq 0, \quad (26)$$

with $\kappa = \text{cond}(M) \leq L^2$;

- (iii) every sublevel set of V_M is forward invariant.

Proof. Since $LK_m < 1$, the inequality $\rho_M < 1$ is equivalent, by (21), to $(1 + LK_m)^2 - 2\alpha\mu_{\text{eff}} + \alpha^2 L^2 K^2 < (1 - LK_m)^2$, that is, to

$$L^2 K^2 \alpha^2 - 2\mu_{\text{eff}} \alpha + 4LK_m < 0.$$

This quadratic in α opens upward and is negative precisely between its roots $\alpha_\pm$; its discriminant condition $\mu_{\text{eff}}^2 > 4L^3 K^2 K_m$ reduces to the second inequality in (24) because $\mu_{\text{eff}} > -K_m K L \geq -2LK\sqrt{LK_m}$ always holds, and then $0 \leq \alpha_- < \alpha_+$. Hence $\rho_M < 1$ for every $\alpha \in (\alpha_-, \alpha_+)$. (The expression under the square root in (21) is nonnegative for every α — otherwise (23), applied to any pair $x \neq y$, would bound a nonnegative quantity by a negative one — so ρ_M is well defined and the squaring above is valid.) By [Lemma 8](#), T is then a ρ_M -contraction on the complete space $(\mathbb{R}^n, \|\cdot\|_{M^{-1}})$, so the Banach fixed-point theorem yields a unique fixed point, which by [Proposition 1](#) is the unique solution $\bar{x}$ of QVI($F, \mathcal{S}$). [Theorem 1](#) and Gronwall's inequality give $\|x(t) - \bar{x}\|_{M^{-1}} \leq \|x(0) - \bar{x}\|_{M^{-1}} e^{-\lambda(1-\rho_M)t}$, and the bounds $\|z\| \leq \sqrt{\|M\|} \|z\|_{M^{-1}}$ and $\|z\|_{M^{-1}} \leq \sqrt{\|M^{-1}\|} \|z\|$, valid for every SPD M , convert this into (26) with prefactor $\sqrt{\|M\|} \|M^{-1}\| = \sqrt{\kappa}$. Forward invariance of the sublevel sets of V_M is immediate from $\frac{dV_M}{dt} \leq 0$. $\square$

Remark 5. The certified exponential rate $\gamma = \lambda(1 - \rho_M)$ scales linearly with λ and depends, through (21), on α, μ, K, K_m , and the matrix bound L . In particular:

- (i) The strong monotonicity modulus μ of F (larger μ increases μ_{eff} , widening the window (25) and decreasing ρ_M).
- (ii) The contraction constant K_m of the translation mapping m (smaller K_m is better: it widens the window and decreases ρ_M).
- (iii) The matrix M through the bound L , sharply the larger of $\|M\|$ and $\|M^{-1}\|$. Note that L is scale-sensitive and is not the condition number of M : for $M = cI$, $\text{cond}(M) = 1$ while the sharp bound is $L = \max(c, c^{-1})$. The condition number enters instead through the Euclidean prefactor in (26).
- (iv) The step-size parameter α (any value in the window (25); the minimizer $\alpha^* = \mu_{\text{eff}} / (L^2 K^2)$ of [Remark 3](#) is the midpoint of the window).

When $m \equiv 0$ (standard VI), the window becomes $(0, \alpha_+)$, where $\alpha_+ = 2\mu_{\text{eff}} / (L^2 K^2)$; with a fixed matrix M , this specializes the certificate to the fixed-set scaled-projection setting studied in [Alshahrani \(2026\)](#), for which the present corollary provides an explicit global guarantee. When additionally $M = I$, we recover the classical exponential stability results for Euclidean projected dynamical systems (with the sharp admissible constant $L = 1$).

Remark 6 (Scale equivalence and certificate normalization). For every $c > 0$, replacing (M, α) by $(cM, \alpha/c)$ leaves the dynamics (11) unchanged: the product αM is unaffected, and $P_{S, (cM)^{-1}} = P_{S, M^{-1}}$, since scaling the objective of (8) by c^{-1} does not change its minimizer. All representatives $(cM, \alpha/c)$ generate the same flow, and the step rule $\alpha = \alpha_{\text{base}} / \|M\|$ used in [Section 5](#) selects precisely the invariant parameterization ($\alpha_{\text{base}} = \alpha \|M\|$ is unchanged along the family). The certificate, by contrast, is representation-dependent: $L = \max(\|M\|, \|M^{-1}\|)$ varies with c , and with it μ_{eff} , ρ_M , and the window (25). Among scalar rescalings, the balanced representative $c^* = \sqrt{\|M^{-1}\| / \|M\|}$ minimizes the admissible bound, attaining $L = \sqrt{\text{cond}(M)}$; indeed it minimizes ρ_M itself along the family (with the paired step α/c). The certificate is therefore sharpest when stated for the balanced representative. No claim about non-scalar preconditioners is intended. Note also that the residual (16) rescales as $r \mapsto c r$ under this reparameterization, so residual-based stopping rules should be referred to a fixed representative.

Remark 7 (Choosing M in practice). Every SPD matrix M is *admissible*: [Assumption 1](#) holds with L the larger of $\|M\|$ and $\|M^{-1}\|$, and $M = I$ always qualifies. No existence question therefore arises. Given M , the certificate constrains the step size α ; the window (25) depends on M through L and may be empty for unfavorable choices. Admissibility is thus weaker than *certification*: the dominance condition (24) is not automatic for every SPD choice. In practice, we recommend the following

deterministic recipe, supported by Experiments 1, 7, and 8. Start with $M = I$. For ill-conditioned problems, try diagonal (Jacobi) scaling of the symmetric part of the operator's linearization. This choice preserves the componentwise projection (Remark 1), but it helps only when the diagonal is heterogeneous: for operators with constant diagonal it is a pure rescaling and leaves the flow unchanged (Remark 6). Adopt a full inverse of the symmetric part only after a condition-number and cost check. Experiment 8 shows that the full inverse can fail under the step rule $\alpha = \alpha_{\text{base}}/\|M\|$ when its conditioning is large. Experiment 7 shows that $M = I$ can win when the nonlinearity distorts the curvature that a preconditioner built from the linear part is designed to correct. Adaptive strategies for choosing M remain an open direction (Section 6).

4.3. Time discretization

The experiments in Section 5 integrate (11) with an adaptive solver to resolve the continuous-time trajectories accurately. As an implementable iterative algorithm with its own guarantee, we take the explicit Euler discretization of (14). Once α lies in the certified window of Corollary 1, the sole additional restriction is $h \leq 1/\lambda$, involving only the user-chosen relaxation rate λ and no problem data.

Proposition 2 (Linear convergence of the forward-Euler scheme). *Let the hypotheses of Corollary 1 hold with $\alpha \in (\alpha_-, \alpha_+)$, so that $\rho_M < 1$, and let $\bar{x}$ be the unique solution of QVI($F, \mathfrak{S}$). Given a step size $h \in (0, 1/\lambda]$, define*

$$x_{k+1} := x_k + \lambda h(T(x_k) - x_k) = (1 - \lambda h)x_k + \lambda h T(x_k) \quad (27)$$

for $k = 0, 1, 2, \dots$. Then for every $x_0 \in \mathbb{R}^n$,

$$\|x_{k+1} - \bar{x}\|_{M^{-1}} \leq q(h) \|x_k - \bar{x}\|_{M^{-1}}, \quad q(h) := 1 - \lambda h(1 - \rho_M) < 1, \quad (28)$$

and hence, with $t_k := kh$,

$$\|x_k - \bar{x}\|_{M^{-1}} \leq q(h)^k \|x_0 - \bar{x}\|_{M^{-1}} \leq e^{-\lambda(1-\rho_M)t_k} \|x_0 - \bar{x}\|_{M^{-1}}.$$

Proof. Since $T(\bar{x}) = \bar{x}$,

$$x_{k+1} - \bar{x} = (1 - \lambda h)(x_k - \bar{x}) + \lambda h(T(x_k) - \bar{x}).$$

For $h \in (0, 1/\lambda]$ the coefficient $1 - \lambda h$ is nonnegative, so the triangle inequality in the M^{-1} -norm and (20) give

$$\|x_{k+1} - \bar{x}\|_{M^{-1}} \leq [(1 - \lambda h) + \lambda h \rho_M] \|x_k - \bar{x}\|_{M^{-1}} = q(h) \|x_k - \bar{x}\|_{M^{-1}}.$$

Iterating — note $q(h) \geq 0$, being a convex combination of 1 and $\rho_M \geq 0$ — and using $1 - s \leq e^{-s}$ with $s = \lambda h(1 - \rho_M)$ yields the final display. $\square$

Remark 8. (i) The factor $q(h)$ is a guaranteed contraction factor per step; the actual progress may be faster. Since $q(h)^k \leq e^{-\lambda(1-\rho_M)t_k}$, the discrete bound is no worse than the decay of Theorem 1 sampled at $t_k = kh$, so the discretization preserves the exponential rate for every $h \in (0, 1/\lambda]$. (ii) The scheme (27) is a Krasnosel'skiĭ–Mann relaxation of T . At $h = 1/\lambda$ it reduces to the fixed-point iteration $x_{k+1} = T(x_k)$ with modulus ρ_M . For $M = I$ this iteration is precisely the projection algorithm for quasi-variational inequalities of Noor (1988) (the case $g = \text{id}$ of his Algorithm 3.1); for general M it is its scaled analogue, a quasi-variational counterpart of the scaled projection step of Bonettini et al. (2009).

4.4. Beyond the certified window

The experiments in Section 5 exhibit convergence far outside the window of Corollary 1. This subsection collects what can be proved about that regime and delimits what remains open.

Proposition 3 (Interior solutions: local stability for every step size). *Suppose Assumptions 1–3 and 5 hold, let $\bar{x}$ solve QVI($F, \mathfrak{S}$), and suppose $\bar{s} := \bar{x} - m(\bar{x}) \in \text{int } S$, with $d := \text{dist}(\bar{s}, \mathbb{R}^n \setminus S) > 0$. Then $F(\bar{x}) = 0$, and*

for every $\alpha > 0$ the solution $\bar{x}$ is exponentially stable, with basin containing the ellipsoid

$$B_\alpha := \{x : \|x - \bar{x}\|_{M^{-1}} < R_\alpha\}, \quad R_\alpha := \frac{d}{C_\alpha \sqrt{\|M\|}}, \quad C_\alpha := 1 + K_m + \alpha \|M\| K,$$

and every trajectory with $x(0) \in B_\alpha$ satisfies

$$\|x(t) - \bar{x}\|_{M^{-1}} \leq \|x(0) - \bar{x}\|_{M^{-1}} e^{-\lambda \alpha \mu \eta t}, \quad \eta := \lambda_{\min}(M).$$

Proof. Since $\bar{s} \in \text{int } S$, the point $\bar{x}$ lies in the interior of $\mathfrak{S}(\bar{x})$, so (2) forces $F(\bar{x}) = 0$; in particular $G(\bar{x}) = \bar{s}$. For $x \in B_\alpha$,

$$\|G(x) - \bar{s}\| \leq (1 + K_m + \alpha \|M\| K) \|x - \bar{x}\| \leq C_\alpha \sqrt{\|M\|} \|x - \bar{x}\|_{M^{-1}} < d,$$

so $G(x) \in \text{int } S$ and $P_{S, M^{-1}}(G(x)) = G(x)$: the projection is inactive, and on B_α the dynamics reduce to $\dot{x} = -\lambda \alpha M F(x)$. There, with $V_M(x) = \frac{1}{2} \|x - \bar{x}\|_{M^{-1}}^2$,

$$\begin{aligned} \frac{dV_M}{dt} &= -\lambda \alpha \langle x - \bar{x}, F(x) - F(\bar{x}) \rangle \\ &\leq -\lambda \alpha \mu \|x - \bar{x}\|^2 \leq -2\lambda \alpha \mu \eta V_M(x), \end{aligned}$$

using $\|z\|^2 \geq \eta \|z\|_{M^{-1}}^2$. Hence V_M is nonincreasing while $x(t) \in B_\alpha$; since B_α is itself a sublevel set of V_M , a first exit time would contradict the continuity of $t \mapsto V_M(x(t))$, so B_α is forward invariant, the reduction remains valid for all $t \geq 0$, and Gronwall's inequality gives the claim. $\square$

The certified rate $\lambda \alpha \mu \eta$ increases linearly in α , consistent with the accelerating convergence in the extreme-step sweep of Section 5.2. The certified radius $R_\alpha = O(1/\alpha)$, however, shrinks as α grows. Proposition 3 therefore predicts fast local decay near the solution and is consistent with the observed terminal behavior; it does not assert that distant initial states reach the basin. Its hypotheses hold, e.g., for the model problem of Section 5 ($\bar{s} = (0.45, 0.27) \in \text{int}[-1, 1]^2$).

Remark 9 (An aligned certificate for translative maps). The worst-case bound $|\langle \delta m, \delta F \rangle| \leq K_m K \|h\|^2$ in the proof of Lemma 8 (where $h = x - y$, $\delta m = m(x) - m(y)$, and $\delta F = F(x) - F(y)$) ignores the alignment of the set motion with the operator. For the translative family $m(x) = \delta x$ with $\delta \in [0, 1]$ and $M = I$, the estimate can be redone exactly: $G(x) - G(y) = (1 - \delta)h - \alpha(F(x) - F(y))$, so $\|T(x) - T(y)\| \leq \rho_\delta \|h\|$ with

$$\rho_\delta := \delta + \sqrt{(1 - \delta)^2 - 2\alpha(1 - \delta)\mu + \alpha^2 K^2}.$$

Here $\rho_\delta < 1$ holds precisely for $0 < \alpha < 2(1 - \delta)\mu/K^2$, with no additional dominance condition. On the model problem of Section 5 ($\mu = 1$, $K = 50$, $\delta = 0.1$), this window is $(0, 7.2 \times 10^{-4})$: nonempty, in contrast with the generic certificate, for which $\mu_{\text{eff}} < 0$ there. Alignment information thus repairs the vacuousness of the worst-case bound on this family, though it still leaves the large-step observations uncertified.

Remark 10 (Compactness yields an ultimate bound, not convergence). Suppose additionally that S is compact with diameter D , and let x_{s_0} be the unique fixed point of the contraction $x \mapsto m(x) + s_0$ for a fixed $s_0 \in S$. Since $T(x) \in m(x) + S$,

$$\|T(x) - x_{s_0}\| \leq K_m \|x - x_{s_0}\| + D,$$

so along trajectories the distance $\rho(t) := \|x(t) - x_{s_0}\|$ satisfies $D^+ \rho \leq \lambda(-(1 - K_m)\rho + D)$ and

$$\limsup_{t \rightarrow \infty} \rho(t) \leq \frac{D}{1 - K_m}.$$

Every trajectory therefore enters every ball around x_{s_0} of radius exceeding $D/(1 - K_m)$ in finite time, for every $\alpha > 0$ and without any monotonicity assumption. This is what compactness of S provably yields: an α -independent ultimate bound, i.e., boundedness of trajectories, not convergence. It explains the absence of divergence in the extreme-step sweep, but cannot be the sole mechanism behind the observed convergence: Experiment 4 converges on the unbounded set $\mathbb{R}_+^2$.

Finally, four estimates account for the conservatism in the certificate ρ_M . First, the cross term $\langle \delta m, \delta F \rangle$ is bounded by its worst-case alignment; [Remark 9](#) shows how much this estimate alone can cost. Second, the projection enters only through plain nonexpansiveness, which discards both firm nonexpansiveness ([Lemma 3](#)) and the saturation of the projection for arguments far outside S . Third, each norm equivalence contributes a factor of L . Fourth, the Cauchy–Schwarz step in [Theorem 1](#) discards all angle information. Within its window, the certificate is a valid lower bound on the decay rate. Outside it, the certificate is silent, and the observed behavior in [Section 5](#) is consistent with mechanisms — alignment, inactivity of the projection near interior solutions, boundedness from compactness — that the worst-case analysis does not capture.

5. Numerical experiments

The goal of this section is to illustrate the behavior of the proposed dynamics (11), to examine the certified convergence behavior, and to explore convergence beyond the sufficient conditions established in [Section 4](#). All experiments are implemented in Julia 1.12. The continuous-time arms use adaptive Runge–Kutta integration via the `OrdinaryDiffEq.jl` package ([Rackauckas & Nie, 2017](#)). `Tsit5` is the default, while the extreme-step stress test of [Section 5.2](#) uses `AutoTsit5` with a `Rosenbrock23` stiff fallback. The discrete baselines of Experiment 9 and the fine-grid $M = A^{-1}$ arm of Experiment 5 at $n \geq 500$ use the explicit iteration of [Section 4.3](#). The projection method used in each experiment is stated in the protocol paragraph below. All computations are performed in double precision on a personal computer equipped with an Intel Core™ i9-9900K processor (3.6 GHz) and 32 GB of RAM. Results are fully reproducible; all source code and scripts are archived on Zenodo and GitHub (see the Data Availability statement). For problems with known analytical solutions (Experiments 1–4, 7, 8), convergence is verified by monitoring both the projection residual $\|r(x(t))\|$ and the distance $\|x(t) - \bar{x}\|$ to the exact solution.

Parameter protocol: unless stated otherwise, all experiments use $\lambda = 1$, residual tolerance 10^{-6} , and the step rule $\alpha = \alpha_{\text{base}}/\|M\|$. Under this rule, the invariant normalized step $\alpha\|M\|$ is identical across the matrices compared within an experiment ([Remark 6](#)). Every compared configuration also uses identical initial-point sets, solver settings, and stopping rule. Any deviation from these shared settings is disclosed where it occurs. In particular, the fine-grid study of Experiment 5 and the baseline comparison of Experiment 9 state each arm’s solver, tolerances, and budget in place.

The values of α_{base} are 0.8 for Experiments 1 and 5, 0.5 for Experiments 3, 4, and 7, and 0.3 for Experiments 6 and 8. Experiment 2 sweeps α directly.

For $M = I$ and diagonal M , projection onto box, orthant, and one-sided sets uses the componentwise clamp. For dense M (Experiments 1–3, 7, and 8), the projection is computed by projected coordinate descent on the box quadratic program. Experiment 6 uses the exact closed-form two-dimensional projection onto the box-with-capacity set. All Experiment-5 statistics use its sparse implementation: the componentwise clamp for diagonal matrices and a KKT-verified primal-dual active-set method for the $M = A^{-1}$ runs.

Throughout, the configuration $M = I$ coincides exactly with the implicit projected dynamical system of [Noor \(2003\)](#) (identify $\rho = \alpha$). Thus, each table’s $M = I$ row doubles as the Euclidean QVI baseline from the literature.

For comparisons across matrices and methods in Experiments 5 and 9, we use the method-independent reference residual

$$R_{\text{ref}}(x) = \frac{1}{\tau} \|x - [m(x) + P_S(x - m(x) - \tau F(x))]\|, \quad (29)$$

with the Euclidean projection and a fixed analytic $\tau = 1/K$ per problem. The native residual $r = (x - T(x))/\alpha$ rescales under $(M, \alpha) \mapsto (cM, \alpha/c)$ ([Remark 6](#)) and therefore cannot compare representations that are scalar rescalings of one another. The cross-matrix tables of Experiments 1, 7,

and 8 report native residuals. No two matrices compared there are scalar multiples of one another. The featured conclusions of those tables are corroborated by the representation-independent distance $\|x - \bar{x}\|$ to the known solution.

Experiments 1–3 use a common two-dimensional affine QVI model problem to explore the effects of the matrix M , step size, and monotonicity in a controlled setting. Experiments 4–8 apply the dynamics to problems from diverse sources: a nonlinear benchmark inspired by the QVI dynamical systems literature ([Noor, 2003](#)), an implicit obstacle problem motivated by contact mechanics ([Elliott, 1987](#); [Noor, 1988](#)), a generalized Nash equilibrium from industrial economics ([Harker, 1991](#)), and synthetic test problems for algorithm validation ([Solodov & Tseng, 1996](#); [Xia & Wang, 2004](#)); Experiment 9 compares against baseline solvers.

Experiments 1–3 share the following two-dimensional affine QVI on $S = [-1, 1]^2$ with operator $F(x) = Qx - b$, where

$$Q = R(\phi)^\top \begin{pmatrix} \sigma & 0 \\ 0 & 1 \end{pmatrix} R(\phi), \quad R(\phi) = \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix},$$

$\sigma = 50$, $\phi = \pi/6$, and the translation mapping $m(x) = \delta x$ with $\delta = 0.1$ (so $K_m = 0.1$). The vector $b = Q\bar{x}$ is chosen so that $F(\bar{x}) = 0$ at a prescribed interior point $\bar{x} = (0.5, 0.3)$, which is verified to be the unique solution of $\text{QVI}(F, \mathfrak{S})$. The operator F is strongly monotone with modulus $\mu = \lambda_{\min}(Q) = 1$ and Lipschitz constant $K = \|Q\| = 50$. Note that the translative structure $\mathfrak{S}(x) = \delta x + S$ allows the QVI to be reduced analytically to a standard VI: the substitution $y = \delta\bar{x} + z$ with $z \in S$ transforms the variational condition into $\langle F(\bar{x}), z - (1 - \delta)\bar{x} \rangle \geq 0$ for all $z \in S$, a VI for the point $(1 - \delta)\bar{x}$. However, this reduction requires knowing the solution; the SPNN-QVI dynamics solve the problem without it.

5.1. Experiment 1: matrix comparison

We compare three choices of M :

- (i) $M = I$ (Euclidean projection);
- (ii) $M = Q^{-1}$ (the inverse of the operator, acting as a perfect preconditioner so that $MF(x) = x - Q^{-1}b$ is isotropic);
- (iii) $M = \text{diag}(1/\sigma, 1)$ (a diagonal approximation of Q^{-1} that captures the eigenvalue scaling but not the rotation ϕ).

For each choice of M , the step size is set to $\alpha = 0.8/\|M\|$ so that the product $\alpha\|M\|$ is constant across all three matrices. We integrate with $\lambda = 1$ and $T = 50$. Fifty random initial points are drawn uniformly on S .

[Fig. 1](#) shows trajectory geometry for eight initial points. Under $M = I$, trajectories spiral toward $\bar{x}$ along the elongated level curves of F . The preconditioned matrix $M = Q^{-1}$ straightens the flow: trajectories approach $\bar{x}$ more directly, reflecting the isotropization of MF . The diagonal matrix $M = \text{diag}(1/\sigma, 1)$ captures the eigenvalue scaling but not the rotation ϕ , producing intermediate behavior.

[Fig. 2](#) compares the convergence profiles from the initial point $x_0 = (1, 1)$. The residual $\|r(x(t))\|$ and the Euclidean distance function $V(x(t)) = \frac{1}{2}\|x(t) - \bar{x}\|^2$ both decay monotonically; the certified quantity in [Theorem 1](#) is the weighted function V_M , so these plots are qualitative illustrations rather than certificate checks. The Euclidean and preconditioned matrices achieve comparable exponential decay rates, while the diagonal matrix converges approximately $5\times$ slower.

[Table 1](#) reports summary statistics over the 50 random initializations. All three matrices achieve 100% convergence (defined by $\|r\|_{\text{final}} \leq 10^{-2}$), but the convergence speed differs markedly. The Euclidean and preconditioned matrices reach residuals of order 10^{-6} – 10^{-7} with a median time-to-tolerance of $t \approx 6$. The diagonal approximation converges to a residual of $\sim 10^{-3}$ with median time $t \approx 29$, reflecting its failure to account for the eigenvector misalignment of Q .

5.2. Experiment 2: step-size sensitivity

The $\rho_M < 1$ condition from [Theorem 1](#) is a sufficient condition for exponential stability. For the model problem with $M = I$, the effective

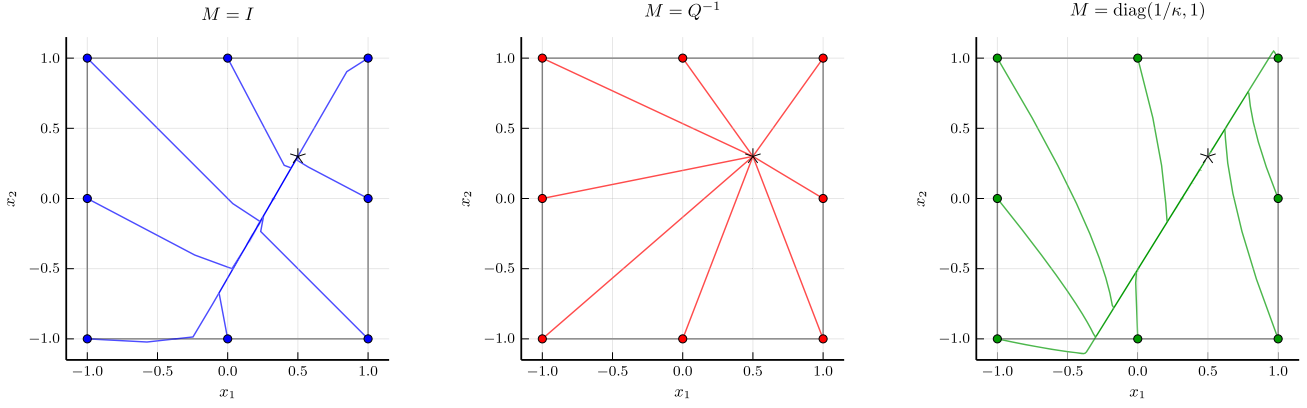


Fig. 1. Phase portraits of the SPNN-QVI dynamics for three choices of M . Eight trajectories from initial points (dots) converge to the solution $\bar{x} = (0.5, 0.3)$ (star) within the constraint box $S = [-1, 1]^2$. Left: $M = I$ (Euclidean); center: $M = Q^{-1}$ (preconditioned); right: $M = \text{diag}(1/\sigma, 1)$ (diagonal approximation).

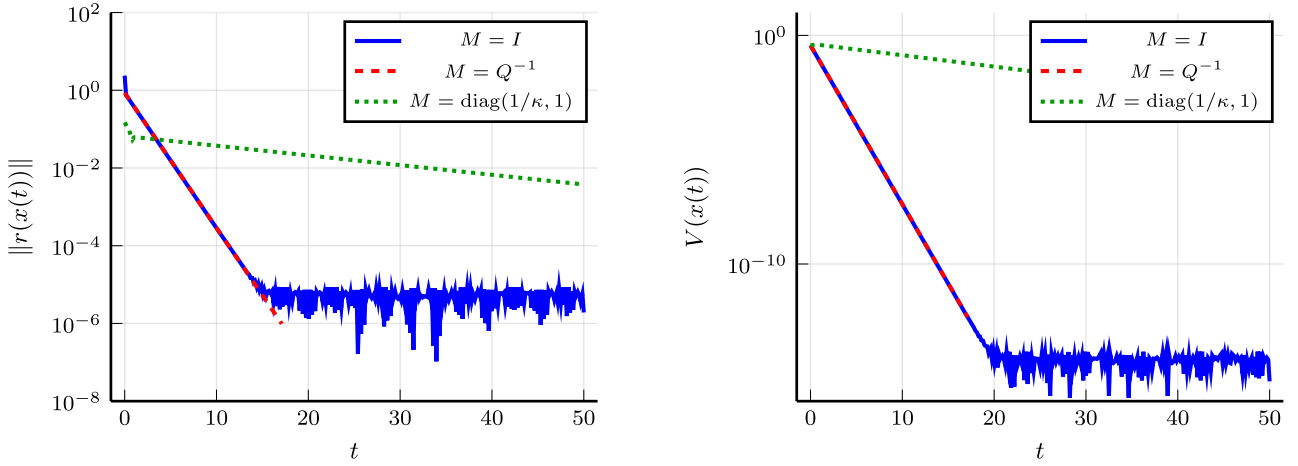


Fig. 2. Convergence from $x_0 = (1, 1)$ for the three matrices: $M = I$ (Euclidean), $M = Q^{-1}$ (preconditioned), $M = \text{diag}(1/\sigma, 1)$ (diagonal). Left: projection residual $\|r(x(t))\|$ on log scale. Right: Euclidean distance function $V(x(t)) = \frac{1}{2} \|x(t) - \bar{x}\|^2$ on log scale. Both quantities decrease monotonically for all three matrices.

Table 1

Benchmark statistics over $N = 50$ random initializations ($\alpha \|M\| = 0.8$, $T = 50$, $\delta = 0.1$). Success is defined by $\|r\|_{\text{final}} \leq 10^{-2}$.

Matrix	Success	$\ r\ _{\text{final}}$	$t(10^{-2})$	$\kappa(M)$
$M = I$	100%	2.4×10^{-6}	6.0	1
$M = Q^{-1}$	100%	7.2×10^{-7}	6.0	50
$M = \text{diag}(1/\sigma, 1)$	100%	2.9×10^{-3}	29.0	50

Table 2

Step-size sweep for $M = I$ ($\delta = 0.1$, $T = 30$). The reference scale of Section 5.2 is $\alpha_{\text{ref}} \approx 0.018$; no step size is certified by the theory for $M = I$ on this problem.

α	αK	$\ r\ _{\text{final}}$	Status
0.01	0.5	5.6×10^{-1}	slow
0.10	5	3.3×10^{-2}	slow
0.50	25	1.6×10^{-6}	converged
1.00	50	7.2×10^{-6}	converged
5.00	250	3.1×10^{-6}	converged
10.00	500	5.5×10^{-6}	converged

monotonicity constant is $\mu_{\text{eff}} = \mu - K_m K = 1 - 5 < 0$, so $\rho_M > 1$ for all $\alpha > 0$ and the theorem does not guarantee convergence. Since no step size is certified here, we use the reference scale $\alpha_{\text{ref}} = (1 - K_m)/(LK) \approx 0.018$ purely as a marker for the sweep. This is the value at which the Lipschitz bound $K_m + \alpha LK$ of the perturbation $m(x) + \alpha MF(x)$ of the identity reaches one; it is neither a necessary nor a sufficient condition for stability. To assess the gap between theory and practice, we sweep α over four orders of magnitude for $M = I$ (Euclidean) and $M = Q^{-1}$ (preconditioned).

Fig. 3 plots the time to reach $\|r\| \leq 10^{-6}$ as a function of α on log-log axes. The vertical dashed line marks $\alpha = 0.018$. Empirically, convergence not only persists but *accelerates monotonically* with increas-

ing α : the convergence time decreases from $t \approx 18$ at $\alpha = 0.8$ to $t \approx 2$ at $\alpha = 10$. For $M = Q^{-1}$, the improvement is even more pronounced, reaching $t(10^{-6}) \approx 2$ already at $\alpha = 5$.

Table 2 provides detailed residuals for $M = I$. Even at $\alpha = 10$ ($\alpha K = 500$), the dynamics converge to $\|r\| \sim 10^{-6}$.

This substantial gap between the certified region and the observed behavior suggests that the projection onto the compact set S provides an implicit regularization mechanism that is not captured by the Lipschitz-based analysis of Section 4.

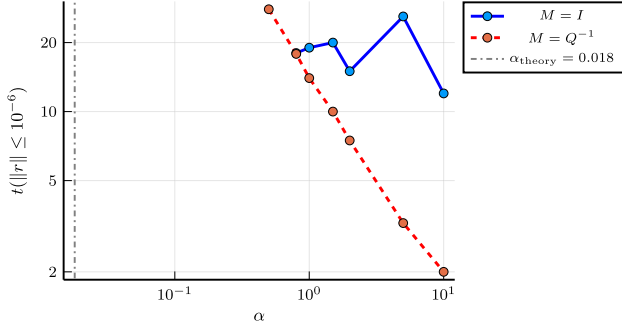


Fig. 3. Time to reach $\|r\| \leq 10^{-6}$ as a function of the step size α (log-log scale) for $M = I$ and $M = Q^{-1}$. The vertical dashed line marks the reference scale $\alpha_{\text{ref}} \approx 0.018$ (see text); the sufficient condition $\rho_M < 1$ from Theorem 1 is not satisfied for any α with $M = I$. Convergence persists and accelerates far beyond this reference scale.

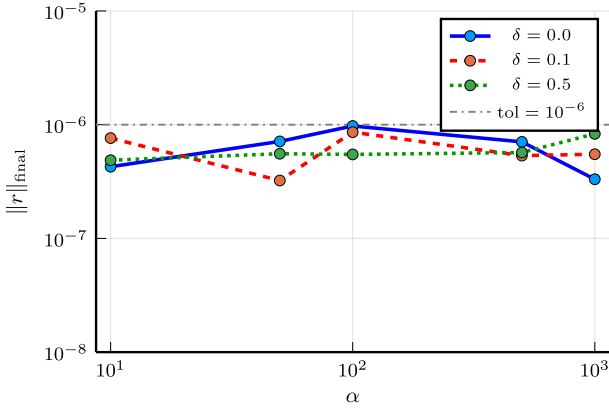


Fig. 4. Final residual $\|r\|_{\text{final}}$ vs. step size α for the two-dimensional model problem with $M = I$ and three values of the contraction constant δ . The horizontal dashed line marks the convergence tolerance 10^{-6} . All tested configurations reached the tolerance, far beyond the certified region of Section 4.

To push further, we test extreme step sizes

$$\alpha \in \{10, 50, 100, 500, 1000\}$$

with contraction constants $\delta \in \{0, 0.1, 0.5\}$ using $M = I$. Fig. 4 plots the final residual as a function of α . All 15 configurations converge to $\|r\| < 10^{-6}$, with convergence time decreasing monotonically from $t \approx 2$ at $\alpha = 10$ to $t \approx 1$ at $\alpha = 1000$ (where $\alpha K = 50,000$). The results are essentially independent of δ , even for $\delta = 0.5$.

We conjecture that, for strongly monotone operators on compact convex sets with the translative QVI structure, the dynamics (11) converge for all $\alpha > 0$. The translation coefficient δ was also swept from 0 to 2.0 (including $K_m > 1$, violating Assumption 2), and all values yield identical convergence.

5.3. Experiment 3: the role of monotonicity and the contraction constant

To understand how the interplay between monotonicity and the contraction constant manifests on a tested grid, we test the dynamics on a family of operators that interpolate between strongly monotone and non-monotone. Specifically, we set $F(x) = (\epsilon I + J)x - b$ where $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ is skew-symmetric, b is chosen so that $\bar{x} = (0.5, 0.3)$ solves the VI on S , and ϵ controls the monotonicity modulus: F is strongly monotone with modulus ϵ when $\epsilon > 0$, purely skew-symmetric (and hence monotone with $\mu = 0$) when $\epsilon = 0$, and non-monotone when $\epsilon < 0$.

Table 3

Residual $\|r\|_{\text{final}}$ as a function of the monotonicity parameter ϵ and the contraction constant $\delta = K_m$. Boldface indicates convergence ($\|r\| < 10^{-6}$).

ϵ	$\delta = 0$	$\delta = 0.1$	$\delta = 0.5$
1.0	9.6×10^{-7}	9.6×10^{-7}	9.6×10^{-7}
0.5	4.9×10^{-6}	4.9×10^{-6}	4.9×10^{-6}
0.1	6.7×10^{-2}	7.9×10^{-2}	9.7×10^{-2}
0.0	4.5×10^{-1}	5.4×10^{-1}	1.1×10^0
-0.1	6.3×10^{-1}	7.5×10^{-1}	1.5×10^0

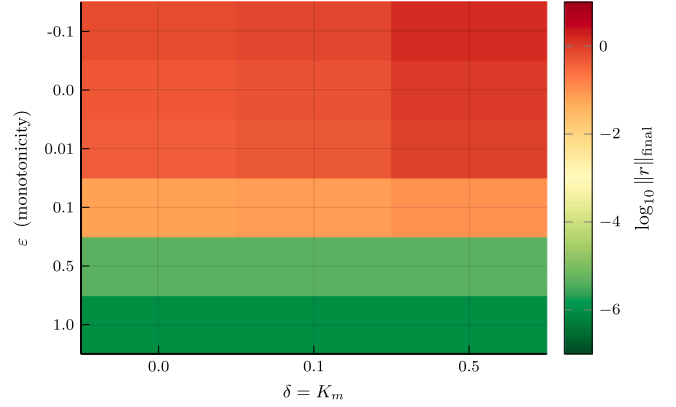


Fig. 5. Heatmap of $\log_{10} \|r\|_{\text{final}}$ as a function of the monotonicity parameter ϵ (vertical axis) and the contraction constant $\delta = K_m$ (horizontal axis). Green regions indicate convergence; red regions indicate stalling. In the tested grid, $\epsilon \geq 0.5$ reached small residuals for every tested δ , while weak or absent monotonicity tolerated only small δ .

Table 3 reports the final residual for several (ϵ, δ) pairs with $M = I$, $\alpha = 0.5$, $T = 50$. Fig. 5 visualizes the same data as a heatmap, making the tested residual pattern immediately apparent.

Two patterns emerge. First, when the monotonicity modulus ϵ is large ($\epsilon \geq 0.5$), small final residuals are reached for every tested δ : the strong monotonicity of F dominates the translation, and the contraction condition $K_m < 1$ is far from binding on the tested grid. Second, when ϵ is small or negative, larger δ systematically worsens the residual. At $\epsilon = 0$ (purely skew), increasing δ from 0 to 0.5 more than doubles the final residual; at $\epsilon = -0.1$ (non-monotone), $\delta = 0.5$ yields $\|r\| = 1.5$, indicating failure. In the tested grid, larger δ degraded convergence exactly when the operator lacked sufficient monotonicity, consistent with the qualitative interplay between the monotonicity modulus μ and the contraction constant K_m predicted by Theorem 1.

We also tested the merely monotone case $F(x) = (x_1, 0)^T$ (monotone with $\mu = 0$; for translation coefficient δ the solution set is $\{0\} \times [-(1 - \delta)^{-1}, (1 - \delta)^{-1}]$). For all tested initial conditions and $\delta \in \{0, 0.1, 0.5\}$, the dynamics converge to the solution set (residual $\sim 10^{-7}$). This is noteworthy because the sufficient condition $\rho_M < 1$ from Theorem 1 is not satisfied under mere monotonicity (see Remark 4), suggesting that the dynamics possess stronger convergence properties than our analysis currently captures.

5.4. Experiment 4: A nonlinear QVI benchmark on the orthant

To test the dynamics on a nonlinear problem with an unbounded set, we consider the following benchmark, adapted from the setting of the Euclidean QVI dynamics of Noor (2003): find $\bar{x} \in \mathbb{R}_+^2$ such that $\langle F(\bar{x}), y - \bar{x} \rangle \geq 0$ for all $y \in \mathfrak{S}(\bar{x})$, where

$$F(x) = \begin{pmatrix} x_1 + x_2 + \sin x_1 \\ x_1 + x_2 + \sin x_2 \end{pmatrix}, \quad m(x) = \frac{1}{8}x, \quad S = \mathbb{R}_+^2.$$

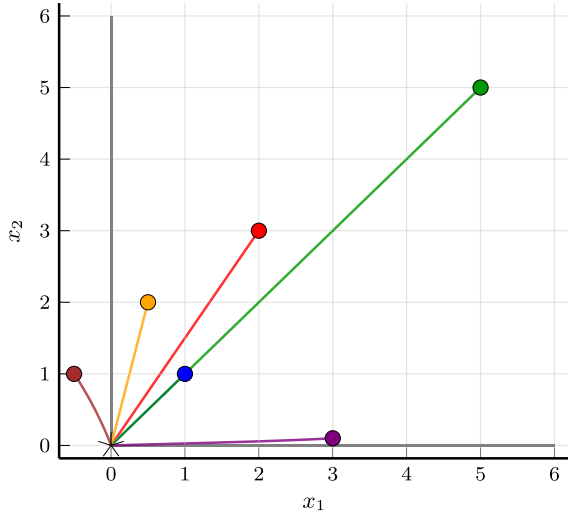


Fig. 6. Trajectories of the SPNN-QVI dynamics for the nonlinear QVI benchmark of Section 5.4. Six initial points (dots) converge to the solution $\bar{x} = (0, 0)$ (star) within the nonnegative orthant $\mathbb{R}_+^2$. The trajectory from $x_0 = (-0.5, 1)$ enters $\mathbb{R}_+^2$ immediately.

The operator F is strongly monotone near the solution: at $\bar{x} = (0, 0)$ the symmetrized Jacobian is $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ with eigenvalues $\{1, 3\}$. It is, however, not globally monotone on $\mathbb{R}_+^2$ (at $u = (\pi, \pi)$ the symmetrized Jacobian has eigenvalue -1), so this benchmark lies outside the global certificate of Section 4. The mapping m is a contraction with $K_m = 1/8$, and the unique solution is $\bar{x} = (0, 0)$. Feasibility $\bar{x} \in \mathcal{G}(\bar{x})$ forces $\bar{x} \geq 0$, and testing (2) with $y = \bar{x}/8 + s$, $s \geq 0$, gives $F(\bar{x}) \geq 0$ and $\frac{7}{8}\langle F(\bar{x}), \bar{x} \rangle = 0$. Since $F_i(x) \geq x_i + \sin x_i \geq 0$ for $x \geq 0$, componentwise complementarity $\bar{x}_i F_i(\bar{x}) = 0$ with $\bar{x}_i > 0$ would force $\bar{x}_i + \bar{x}_2 + \sin \bar{x}_i = 0$, impossible for $\bar{x} \neq 0$; hence $\bar{x} = 0$. We run the dynamics with $M = I$: precisely the implicit projected dynamical system of Noor (2003), applied to this benchmark.

Fig. 6 shows the phase portrait. From six initial points (including $x_0 = (-0.5, 1)$, which starts outside $\mathbb{R}_+^2$), all runs converge to $\bar{x} = (0, 0)$ with $\|x - \bar{x}\| < 6 \times 10^{-9}$ and $\|r\| < 10^{-8}$. The median convergence times are $t(10^{-2}) \approx 7$ and $t(10^{-6}) \approx 17$, with $t(10^{-8}) \approx 23$. The dynamics handle the unbounded constraint set $\mathbb{R}_+^2$ and the nonlinear operator (which includes $\sin x_i$) without difficulty.

5.5. Experiment 5: implicit obstacle problem

Quasi-variational inequalities originated in obstacle and free boundary problems (Bensoussan & Lions, 1982; Elliott, 1987), and translative QVIs are a classical model class in this context (Baiocchi & Capelo, 1984; Noor, 1988). Motivated by these works, we construct an illustrative responsive-substrate model: a membrane constrained to lie above a substrate whose height responds to the membrane's displacement, giving the translative structure $\mathcal{G}(x) = m(x) + S$. We discretize such a problem on $[0, 1]$ with $n = 20$ interior grid points ($h = 1/(n + 1)$).

The stiffness matrix is the standard tridiagonal

$$A = \frac{1}{h^2} \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & \ddots & & \\ & \ddots & \ddots & \ddots & \\ & & & -1 & 2 \end{pmatrix} \in \mathbb{R}^{n \times n},$$

with uniform load $f_i = 1$, and the base obstacle profile is $\psi_{0,i} = 0.2 \sin(\pi x_i)$ where $x_i = ih$. The constraint set is $S = \{x \in \mathbb{R}^n : x \geq \psi_0\}$. The translation $m(x) = \delta \langle x \rangle \mathbf{1}_n$ (where $\langle x \rangle := \frac{1}{n} \sum_{i=1}^n x_i$ is the mean displacement) models the substrate response: the obstacle height shifts uniformly by a fraction $\delta = 0.1$ of the average membrane displacement, an

illustrative model of a globally compliant elastic foundation. The Lipschitz constant of m is $K_m = \delta < 1$, attained at perturbations proportional to $\mathbf{1}_n$.

The stiffness matrix has $\kappa(A) \approx 178$. Since $\text{diag}(A)$ is constant, the Jacobi matrix $\text{diag}(A)^{-1} = (h^2/2)I$ is a scalar multiple of the identity and generates *exactly the same flow* as $M = I$ (Remark 6). It is therefore reported as a separate control run in the text rather than as a row of Table 4, and the substantive comparison is between the unpreconditioned flow and the operator preconditioner $M = A^{-1}$. Because the native residual $r = (x - T(x))/\alpha$ is representation-dependent, all cross-matrix statements below use the reference residual R_{ref} defined in (29). Fig. 7 shows the converged membrane profile alongside the base obstacle ψ_0 , and Table 4 reports the convergence statistics. A dedicated step-size sweep for $M = I$ against the *absolute* threshold $R_{\text{ref}} \leq 10^{-6}$ uses 22 values of α over five decades including the protocol value, four starts, horizon $T = 200$, and reference-residual stopping at accepted integration end-points. Only two tested grid points, $\alpha \approx 1.8 \times 10^{-3}$ and 3.2×10^{-3} , reach that threshold from all four starts, with worst-case hit times $t \approx 183$ and 106. The remaining tested steps — including the protocol value, where the worst final reference residual is 8×10^{-4} — do not reach it within the integration budget. Thus at $n = 20$ the unpreconditioned flow converges to this accuracy only inside a narrow, problem-specific step window.

Table 4 reports the statistics at $n = 20$ under a common protocol: every arm shares the tight tolerance pair $10^{-12}/10^{-10}$. Under the relative work criterion of the fine-grid study below, both configurations meet the target from all four starts in comparable model time. At this dimension the step size, not the matrix, is the binding factor: the sweep above shows the unpreconditioned flow reaches the *absolute* threshold only inside a narrow tuned step window. The Jacobi arm, run separately as a scale-equivalence control, reproduces all four $M = I$ hit decisions and agrees at the reported model-time precision. The two representations generate the identical flow in exact arithmetic (Remark 6). Their raw adaptive-integration telemetry and terminal residuals differ slightly at the solver-tolerance level, while the common residual, settings, and reported outcomes confirm the predicted scale equivalence. (In an earlier pilot with per-arm tolerances the same two arms showed pass/fail flips at the marginal threshold — cross-representation comparisons need both a common residual and common settings.) This experiment, together with the fine-grid study below, demonstrates that for ill-conditioned problems arising from PDE discretizations, the choice of configuration — the $M = A^{-1}$ implementation against the unpreconditioned one, each under its stated solver, tolerances, and budget — determines whether the declared work target remains attainable as the discretization is refined, without per-problem step-size tuning.

To examine conditioning at scale, we repeat the experiment on finer grids with a sparse implementation. The stiffness matrix is stored in sparse form, and diagonal matrices use the componentwise clamp. For $M = A^{-1}$, the matrix is applied through a sparse Cholesky factorization. At every evaluation, the metric projection, a tridiagonal obstacle-type quadratic program, is solved by a primal-dual active-set method. Each solve is verified against its KKT system to a scale-aware tolerance.

At fine grids ($n \geq 500$), the $M = A^{-1}$ configuration is advanced by the discrete Krasnosel'skiĭ–Mann scheme of Section 4.3, with relaxation step $h = 0.1$. Adaptive integration of the projected flow is impractical there because the error controller stalls at active-set kinks. Each configuration's solver is reported with its results. The discrete scheme is bounded by its iteration budget rather than by the common horizon T . Consequently, model times are compared only within a solver.

Each projection performs at most $O(n)$ reduced tridiagonal solves, each of $O(n)$ cost. Thus, a single projection is bounded by $O(n^2)$ work in the worst case. The run manifest written by the archived script records the discrete scheme's iteration budget and relaxation step (see the Data Availability statement).

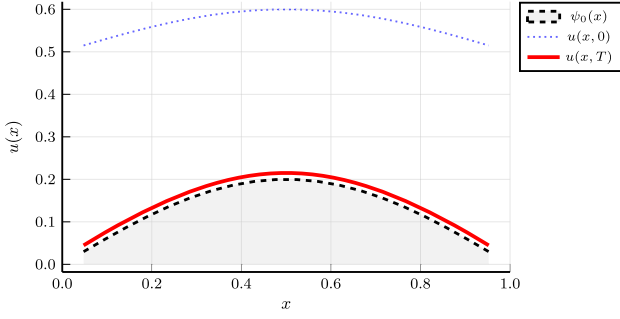


Fig. 7. Membrane displacement $u(x)$ for the implicit obstacle QVI ($n = 20$, $M = A^{-1}$). The dashed curve shows the base obstacle $\psi_0(x) = 0.2 \sin(\pi x)$; the dotted curve is the initial condition; the solid curve is the converged solution. The membrane settles above the obstacle, with contact near the obstacle peak.

Table 4

Implicit obstacle QVI ($n = 20$, $\delta = 0.1$, $T = 200$, protocol step $\alpha_{\text{base}} = 0.8$; all arms share the tolerance pair $10^{-12}/10^{-10}$). Success is meeting the relative work target of the fine-grid study (the reference residual (29) reduced to 10^{-6} of its initial value, checked at accepted integration endpoints); the third column is the median final relative reduction over the four starts, and t_{hit} the median model time of the successful starts. The Jacobi matrix $\text{diag}(A)^{-1}$ generates the same flow as $M = I$ (Remark 6) and is therefore not a separate row; see the text for the control run.

Matrix	Rel. hits	$R_{\text{ref}}/R_{\text{ref}}(x_0)$	t_{hit}
$M = I$	4/4	1.0×10^{-6}	17.7
$M = A^{-1}$	4/4	7.3×10^{-7}	18.0

Success is assessed by a relative residual-reduction criterion: the reference residual (29) must be reduced to 10^{-6} of its initial value. This criterion was adopted after a pilot showed that the originally declared fixed absolute threshold lay below the attainable integrator accuracy at large n , and it was locked before the production runs. It is a matched-start work criterion, not a claim of common physical accuracy across grids. The per-run output of the archived script also records the absolute rule.

Unpreconditioned runs at fine grids are budget-censored. Their admissible explicit step scales as $1/(\alpha \lambda_{\max}(A))$, and the recorded integration horizon quantifies that cost blow-up. At the largest dimension, the unpreconditioned arm is run from a single demonstration start, while $M = A^{-1}$ runs all four.

Where Tsit5 is used, the $M = A^{-1}$ arm uses tolerances $10^{-12}/10^{-10}$ for $n \leq 100$. The unpreconditioned arms use tolerances $10^{-8}/10^{-6}$ for $n \geq 100$. The tighter tolerances used for the preconditioned arm make its task harder, not easier. At $n \geq 500$, the $M = A^{-1}$ arm uses the discrete scheme, to which integrator tolerances do not apply. Both solvers carry a step or iteration budget of 2×10^5 . The wall caps are 40 s for the censored unpreconditioned arms and 900 s for the $M = A^{-1}$ arm.

Table 5 summarizes the outcome. The configuration $M = A^{-1}$ meets the relative work target from all four starts at every tested dimension. The largest verified dimension is $n = 10^4$, where $\kappa(A) \approx 4.1 \times 10^7$. Its model hit time is essentially dimension-independent, with median $t \approx 16$ –19 across three decades of n . The values for $n \geq 500$ arise from the discrete scheme, whose model time is the first-order clock of the same flow, so this range is indicative rather than a within-solver comparison. Wall cost grows with the projection work, from 0.07 s per run at $n = 20$ to about 292 s at $n = 10^4$. The unpreconditioned configuration degrades structurally. At $n = 100$, it meets the target from one of four starts within its step budget. From $n = 500$ onward, its admissible explicit step collapses, so the integration is censored at ever-earlier model

Table 5

Fine-grid obstacle QVI: dimension and conditioning scaling. Hits count the starts that meet the relative work target. There are four starts, except that the unpreconditioned arm uses a single demonstration start at $n = 10^4$. The quantity t_{hit} is the median model hit time of successful starts. The wall column gives the median wall-clock seconds per run for the $M = A^{-1}$ arm. That arm uses Tsit5 at $n \leq 100$ and the discrete scheme of Section 4.3, with $h = 0.1$, at $n \geq 500$. Model times are comparable only within a solver. For $M = I$, t_{end} is the median censored model endpoint (step-budget-censored at $n = 100$ –500; wall-censored at 40 s beyond). The Jacobi arm is omitted (identical flow to $M = I$; Remark 6); per-arm tolerances are stated in the text.

n	$\kappa(A)$	$M = A^{-1}$			$M = I$	
		Hits	t_{hit}	wall (s)	Hits	t_{end}
20	1.8×10^2	4/4	18.0	0.07	4/4	—
100	4.1×10^3	4/4	18.5	0.56	1/4	31.0
500	1.0×10^5	4/4	18.4	1.3	0/4	2.0
1000	4.1×10^5	4/4	18.8	3.4	0/4	0.21
10^4	4.1×10^7	4/4	16.3	292	0/1	0.00

times ($t_{\text{end}} \approx 2.0, 0.21$, and 0.00 at $n = 500, 1000, 10^4$). These are observations about the stated implementations under their stated budgets; no cross-solver speedup is claimed.

5.6. Experiment 6: generalized Nash equilibrium

Generalized Nash equilibrium problems (GNEPs), in which each player's feasible set depends on the rivals' strategies, are a canonical source of quasi-variational inequalities (Facchinei & Kanzow, 2010; Harker, 1991). We consider a Cournot duopoly: two firms choose production quantities $x_1, x_2 \geq 0$ facing inverse demand $p(Q) = d - \lambda_p - \rho Q$ (where $Q = x_1 + x_2$) and asymmetric marginal costs $c_1 = 1, c_2 = 2$, with parameters $d = 20, \lambda_p = 4, \rho = 1$.

The pseudo-gradient operator is

$$F(x) = \begin{pmatrix} 2\rho x_1 + \rho x_2 - (d - \lambda_p - c_1) \\ \rho x_1 + 2\rho x_2 - (d - \lambda_p - c_2) \end{pmatrix},$$

with individual box constraints $x_i \in [0, 10]$ and a shared capacity constraint $x_1 + x_2 \leq 12$. The translation $m(x) = \delta x$, with $\delta = 0.1$, models committed production. A fraction δ of each firm's current output is pre-committed, for instance under bilateral contracts. The set S is interpreted as *residual-market quotas*: per-firm and aggregate limits on the uncommitted output $x - m(x)$. Consequently, the feasible set $\mathcal{E}(x) = m(x) + S$ shifts with the state, which gives the quasi-variational structure. The example is a stylized jointly constrained Cournot model rather than a player-wise generalized Nash derivation. The operator F is strongly monotone (the Jacobian has eigenvalues ρ and 3ρ).

From ten initial points (with $M = I$), the dynamics converge to the unique Nash equilibrium $\bar{x} \approx (5.33, 4.33)$ with maximum pairwise spread $\sim 3.0 \times 10^{-6}$ among final iterates. The equilibrium is economically intuitive: firm 1, having lower marginal cost, produces more. Six of ten starts reach $\|r\| < 10^{-6}$; the remaining four are at $\|r\| \approx 1.3$ – 1.7×10^{-6} , barely above the tolerance.

5.7. Experiment 7: nonlinear monotone operator

All preceding experiments use affine or mildly nonlinear operators. To test the framework on a genuinely nonlinear problem, we consider the operator

$$F(x) = Qx + \rho \varphi(x) + q, \quad \varphi_i(x) = \arctan(x_i - 2),$$

on $S = [0, 5]^5$ with $m(x) = 0.1x$, following the test problem structure used in projection neural network studies (Hu & Wang, 2006; Xia & Wang, 2004). Here $Q \in \mathbb{R}^{5 \times 5}$ is symmetric positive definite with eigenvalues (10, 5, 2, 1, 0.5) and a fixed random eigenbasis, $\rho = 1$ controls the

Table 6

Nonlinear monotone QVI ($n = 5$, $\rho = 1$, $\delta = 0.1$, $T = 80$; convergence: $\|r\|_{\text{final}} \leq 10^{-6}$). Convergence from six initial points.

Matrix	Conv.	$\ r\ _{\text{final}}$	$t(10^{-6})$	$\ x - \bar{x}\ $
$M = I$	6/6	9.2×10^{-7}	20.2	6.5×10^{-7}
$M = Q^{-1}$	6/6	9.6×10^{-7}	54.3	9.1×10^{-7}

Table 7

Scaling study: convergence statistics over 10 random starts per configuration ($\delta = 0.1$, $T = 100$; convergence: $\|r\|_{\text{final}} \leq 10^{-6}$). The diagonal preconditioner is the only matrix that achieves near-convergence at all dimensions.

n	Matrix	Conv.	$\ r\ _{\text{final}}$	$t(10^{-2})$	$\kappa(M)$
10	$M = I$	0/10	6.8×10^{-5}	16.5	1
10	$M = D^{-1}$	0/10	1.1×10^{-2}	94.5	158
10	$M = \text{diag}^{-1}$	2/10	1.5×10^{-6}	38.0	8
20	$M = I$	0/10	4.9×10^{-5}	14.0	1
20	$M = D^{-1}$	0/10	5.0×10^{-1}	∞	464
20	$M = \text{diag}^{-1}$	0/10	2.2×10^{-6}	17.0	4
50	$M = I$	0/10	3.3×10^{-4}	10.0	1
50	$M = D^{-1}$	0/10	4.1×10^0	∞	1725
50	$M = \text{diag}^{-1}$	0/10	4.1×10^{-6}	9.0	2

nonlinearity strength, and q is chosen so that $\bar{x} = (1, 2, 3, 2, 1)$ is the solution.

The componentwise nonlinearity φ is monotone ($\varphi'_i \geq 0$), so F is strongly monotone. However, the nonlinear term changes the effective curvature near $\bar{x}$, which affects the optimal choice of M .

Table 6 compares $M = I$ with $M = Q^{-1}$ over six initial conditions. Both matrices achieve 100% convergence, but $M = I$ is approximately $2.5\times$ faster (median $t(10^{-6}) \approx 20$ vs. 54). This is because Q^{-1} isotropizes only the linear part of F ; the nonlinear term $\rho \varphi(x)$ introduces curvature that is not captured by the preconditioner. This illustrates that the “best” matrix M depends on the full problem structure, not just the leading linear term.

5.8. Experiment 8: higher-dimensional scaling

To assess how the dynamics scale with problem dimension, we generate random strongly monotone QVIs of dimension $n \in \{10, 20, 50\}$ following the standard construction of Solodov and Tseng (1996). For each n , we form $Q = D + S$ where $D = N^T N + 0.1I$ (symmetric positive definite, N with i.i.d. standard normal entries) and S is skew-symmetric, making Q positive definite but asymmetric. The operator is $F(x) = Qx + q$ on $S = [0, 10]^n$ with $m(x) = 0.1x$ and $q = -Q\bar{x}$ for a known interior solution $\bar{x}$. A fixed random seed ensures reproducibility.

We compare three matrices: $M = I$ (Euclidean), $M = D^{-1}$ (symmetric-part inverse), and $M = \text{diag}(Q)^{-1}$ (diagonal scaling). For each (n, M) pair, ten random initial points are drawn uniformly on S .

Table 7 reveals three distinct behaviors over the tested horizon and tolerance. First, $M = I$ produces residuals that worsen with dimension (10^{-5} at $n = 10$, 10^{-4} at $n = 50$) and does not reach 10^{-6} . Second, the full symmetric-part inverse $M = D^{-1}$ degrades sharply as n grows: its condition number increases from 158 to 1725, and under the step rule $\alpha = \alpha_{\text{base}}/\|M\|$ the effective step $\|\alpha M\| = \alpha_{\text{base}}$ is unchanged, but its weakest scaled direction $\lambda_{\min}(\alpha M) = \alpha_{\text{base}}/\kappa(M)$ shrinks with the conditioning. At $n = 50$, the residual exceeds 4. Third, the diagonal preconditioner $M = \text{diag}(Q)^{-1}$ maintains near-convergence across all dimensions: its condition number stays small ($\kappa \leq 8$), its weakest scaled direction stays bounded away from zero, and the median residual is consistently $\sim 10^{-6}$.

This experiment highlights an important practical lesson: a theoretically “optimal” preconditioner can be counterproductive if its condi-

Table 8

Baseline comparison (Experiment 9): held-out success (hits/starts), median end-to-end solve time, median complete map evaluations (one F evaluation and one metric projection each), and median final reference residual (29) per cell. Every cell terminated by reaching the target (no timeouts). Stored times below the millisecond resolution are reported as < 1 ms; setup (construction and factorization), reported separately, is 1.3/0.7/0.8 s for the three problems. Locked step parameters and per-run data are regenerated, together with a run manifest, by the archived comparison script (see the Data Availability statement).

Problem	Method	Hits	t_{med}	Map evals	R_{ref}
Obstacle ($n = 20$)	SPNN, $M = A^{-1}$	7/7	15 ms	297	6.0×10^{-7}
	Noor PDS ($M = I$)	7/7	< 1 ms	1809	8.4×10^{-7}
	Fixed point ($M = I$)	7/7	< 1 ms	1530	9.8×10^{-7}
	Fixed point, $M = A^{-1}$	7/7	< 1 ms	106	9.4×10^{-7}
Nonlinear ($n = 5$)	SPNN, diag. M	7/7	< 1 ms	1683	9.6×10^{-7}
	Noor PDS ($M = I$)	7/7	< 1 ms	699	9.4×10^{-7}
	Fixed point ($M = I$)	7/7	< 1 ms	173	9.5×10^{-7}
	Fixed point, diag. M	7/7	< 1 ms	2762	1.0×10^{-6}
Random ($n = 50$)	SPNN, diag. M	7/7	11 ms	3207	9.9×10^{-7}
	Noor PDS ($M = I$)	7/7	42 ms	2355	9.9×10^{-7}
	Fixed point ($M = I$)	7/7	12 ms	592	9.9×10^{-7}
	Fixed point, diag. M	7/7	4 ms	1536	9.9×10^{-7}

tion number is high, because directional attenuation amplifies the ill-conditioning. In contrast, a simple diagonal scaling that keeps $\kappa(M)$ small can outperform the full inverse, keeping all scaled directions active.

5.9. Experiment 9: comparison with baseline solvers

Finally, we compare the proposed dynamics against the baselines suggested by the QVI literature, under the method-independent reference residual (29) and the fairness protocol of the Section 5 preamble: (i) the scaled SPNN dynamics with the deterministic recipe matrix of Remark 7 (integrated by Tsit5); (ii) the same dynamics with $M = I$, i.e., precisely the implicit projected dynamical system of Noor (2003); (iii) the discrete fixed-point projection iteration $x_{k+1} = T(x_k)$ with $M = I$, i.e., the classical algorithm of Noor (1988) (Proposition 2 with $h = 1/\lambda$); and (iv) its scaled variant. For each method, the step parameter α_{base} is tuned on the same 21-point logarithmic grid over $[10^{-4}, 10]$, using dedicated pilot starts, and is then locked. Performance is reported on held-out starts as the median over runs. Per-run solve time is measured end-to-end, including the residual checks but excluding the separately reported setup time. Function-evaluation and projection counts are recorded separately from diagnostics.

Each method runs under a formulation-specific internal budget. The continuous arms use integration horizon $T = 500$, step budget 2×10^5 , and tolerances $10^{-8}/10^{-6}$. The discrete iterations use an outer-iteration cap of 2×10^5 . All methods also use a common 60 s wall-clock check, polled at completed steps. Because this wall check is not hard-enforced, success rates are not compared as outcomes under a single binding budget. Each cell’s terminating cause is reported.

Timeouts are censored failures and are never averaged into times. Several fast cells are below the millisecond resolution of the stored wall times and are reported as “ < 1 ms.” Operation counts, rather than relative differences among sub-millisecond timings, carry the work comparison.

Three representative problems are used: the implicit obstacle problem (Experiment 5), the nonlinear monotone problem (Experiment 7), and the high-dimensional problem (Experiment 8, $n = 50$).

Table 8 reports the outcome. Every method reaches the reference target from every held-out start on all three problems under its locked step (twelve of twelve cells at 7/7, zero timeouts). At these dimensions all four solvers are effective, and there is no success-rate separation un-

der the stated budgets. The informative contrast is in work. On the ill-conditioned obstacle problem the scaled arms need six- to fourteen-fold fewer complete map evaluations than their Euclidean counterparts (297 vs. 1809 for the continuous dynamics; 106 vs. 1530 for the discrete iteration). The benefit of the matrix M thus transfers from the continuous flow to the discrete scheme, reinforcing rather than replacing the continuous-time contribution. On the nonlinear problem the ordering reverses for the discrete arms (173 vs. 2762), consistent with Experiment 7, where $M = I$ outperforms the scaled matrix; on the random problem the counts lie within a factor of about 5.4 across methods. Median end-to-end times are milliseconds throughout, and we do not read rankings into them. For most cells several step candidates succeed in the tuning phase, and among succeeding candidates the lock selection rests on timing differences at or near the stored millisecond resolution. Operation counts under the recorded locks, not wall times, therefore carry the comparison. A paired tolerance-tightening audit reproduced every per-start success and termination classification (84 of 84).

6. Concluding remarks

We have proposed and analyzed a scaled projection neural network for quasi-variational inequalities, in which a fixed symmetric positive definite matrix M defines the geometry of the projection and the constraint set $\mathfrak{S}(x) = m(x) + S$ varies with the state through a contraction mapping m .

The main results establish: (i) an exact correspondence between equilibria of the dynamics and solutions of the QVI; (ii) well-posedness (existence and uniqueness of trajectories) under Lipschitz regularity of F and m ; (iii) Lyapunov stability via an M^{-1} -weighted Lyapunov function with an explicit contraction parameter ρ_M that captures the interplay between the matrix bound L , the contraction constant of m , and the operator's monotonicity properties; and (iv) global exponential convergence on an explicit step-size window, under the quantitative dominance conditions of (24), with a rate determined by λ , α , μ , K , K_m , and the matrix bound L , together with a forward-Euler discretization preserving the rate.

The framework provides a natural continuous-time interpretation of scaled projection methods for quasi-variational inequalities. The matrix M acts as a fixed preconditioner that reshapes the geometry of the projected flow without altering the solution set. At the same time, the translation structure $\mathfrak{S}(x) = m(x) + S$ allows the constraint set to move with the state in a controlled manner, provided m is a contraction.

Several directions remain open. First, the current stability analysis requires strong monotonicity to achieve $\rho_M < 1$; under mere monotonicity, the cross terms from the moving constraint set prevent the weighted Lyapunov analysis from guaranteeing stability. Extending the convergence theory to *monotone* or *pseudomonotone* operators (as in Hu & Wang, 2006; Noor, 2003), perhaps by exploiting the variational characterization of the projection directly rather than norm bounds, would broaden the applicability. Second, allowing M to be *state-dependent* $M(x)$ simultaneously with the moving constraint set $\mathfrak{S}(x)$ would unify our framework with that of Alshahrani (2026) into a fully coupled setting. Third, Proposition 2 covers the explicit Euler discretization inside the certified step-size window; convergence guarantees for higher-order or implicit discretizations, and for discrete schemes outside the contraction regime, remain open. Fourth, investigating adaptive or data-driven strategies for choosing M based on problem structure remains a promising direction. Finally, the numerical experiments reveal a substantial gap between the certified sufficient conditions and the observed convergence (convergence persists for step sizes exceeding the reference scale of Section 5.2 by a factor of 10^4). Section 4.4 delimits what interior structure, alignment, and compactness provably yield in that regime. Establishing a global guarantee for large step sizes remains an important open problem. Resolving it would close the remaining gap and give convergence guarantees that are tighter in practice.

CRedit authorship contribution statement

Mohammed Alshahrani: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Qamrul Hasan Ansari:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) in order to assist with prose editing, verifying LaTeX formatting, and checking internal consistency of cross-references and notation. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All source code and scripts that reproduce the numerical experiments reported in this paper are archived on Zenodo at <https://doi.org/10.5281/zenodo.21521187> (version 1.1.0), with a development mirror maintained at <https://github.com/mmogib/SPNN-QVI>. No external datasets were used.

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